SORTIE: Fair Vehicle Routing When Route Durations Are Random.

Ex-post ordered fairness, rank prices and risk-consistent tours

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Abstract

Fair vehicle routing balances drivers' workloads with ordered measures of route lengths, such as the range or the Gini mean difference. The recent exact approaches, which also require every route to be a shortest tour of its customers, assume deterministic travel times. Drivers, however, experience realised durations. We study fair routing when arc travel times are random, with a fleet-wide day factor and arc-level noise. We distinguish ex-ante fairness (of expected workloads) from ex-post fairness (the expected unfairness of the realised day, sorted day by day) and show that every deterministic fair plan overstates ex-post fairness, with an explicit bound in terms of the routes' quadratic lengths. We prove that multiplicative common shocks leave expected ordered fairness unchanged. We then give an exact, sampling-free formula for the expected value of any ordered measure of independent route durations, a Poisson–binomial order-statistic integral that also handles the negative weights of dispersion measures, and derive from it *rank prices*: the derivative of expected unfairness with respect to a shift of each route's duration. Negative rank prices identify exactly the routes a fairness objective would like to lengthen. Under uncertainty even the min-max measure can prefer a tour that is not the shortest, so route consistency must be imposed explicitly; we define it through each driver's own mean–standard deviation and prove that such tours are supported Pareto tours, computable by parametric Held–Karp. The resulting bilevel heuristic, SORTIE, screens all relocate and swap moves with the rank prices. On instances derived from the CVRPLIB benchmark used by the deterministic literature, deterministic fair plans understate their realised unfairness by 8–27%, while the integral agrees with Monte Carlo to about 0.1% at under a millisecond per plan. In seventeen experiments on two benchmark families, optimising ex-post fairness improves realised fairness by about 1% at the standard budget, 2% on a second family of twelve CVRPLIB instances (better on all twelve), and up to 7% at generous budgets, where it also uses less of the budget. It significantly beats sample average approximation with up to 1000 scenarios, and a risk-adjusted deterministic baseline, which does no better than ignoring uncertainty. Rank-price screening ranks moves with a Spearman correlation of about 0.9 with their exact effect, and without the lower level nearly all range-optimised plans buy fairness with detours.

Keywords: vehicle routing; fairness; stochastic travel times; ordered weighted averaging; order statistics; bilevel optimisation.

1 Introduction

Routing plans that minimise total distance often give some drivers much longer days than others. Unequal workloads hurt driver satisfaction and retention and can be seen as unfair [Matl et al.,

2018]. The fairness literature in routing therefore optimises a *fairness measure* of the vector of route workloads: the longest route, the range, the Gini mean difference (GMD), the variance or a mean absolute deviation [Matl et al., 2018, Karsu and Morton, 2015, Bektaş and Letchford, 2020]. Ordered measures, which weight the sorted workloads [Yager, 1988, Ogryczak and Śliwiński, 2003, Nickel and Puerto, 2005], cover all of these in one framework [Blanco et al., 2026].

Non-monotone measures such as the range or the GMD create a well-known pathology: an optimiser can lower the measure by making a light route *longer*, giving a driver a needless detour [Matl et al., 2018]. van Rossum et al. [2026a,b] and Ljubić et al. [2026] close this loophole exactly, by requiring every route to be a shortest (TSP-optimal) tour of its customers. Ljubić et al. [2026] write the problem as a bilevel programme in which a planner assigns customers and each driver solves a TSP, and solve instances with up to 35 customers to optimality for seven measures.

All of these works assume deterministic travel times. A driver's day, however, is a realised duration: congestion, weather, parking and service-time variability differ between routes and between days. Fairness is then a statement about random variables, and three questions arise that the deterministic theory does not ask.

1. *What should be balanced?* The expected workloads (ex-ante fairness), or the workloads drivers actually experience, sorted day by day (ex-post fairness)? The identity of the worst-off driver changes from day to day, and the two notions can disagree.

2. *How can ex-post fairness be optimised?* Sample-average approximation (SAA) is the default, but sorting per scenario makes the objective noisy and its gradient uninformative.

3. *What is a consistent route?* Under random durations "the shortest tour" is not obviously what a driver would choose, and we show that monotone measures no longer protect drivers from detours.

Contributions.

- **Model.** We formulate fair routing with random travel times (a fleet-wide day factor times arc-level noise) as a bilevel problem with an ex-post ordered fairness objective, an expected-cost budget and per-driver risk-consistent tours. With no noise it reduces to the X-CVRP-TSP of Ljubić et al. [2026].

- **Fairness illusion (Proposition 5).** For every ordered measure with nondecreasing weights, ex-ante fairness underestimates ex-post unfairness, by at most $\|\lambda\|_1 v(\sum_k W_k)^{1/2}$, where W_k is the route's quadratic length. Common multiplicative shocks leave the expected value of every ordered measure unchanged (Proposition 2); they do inflate the variance measure and the tail of unfairness.

- **PBOI (Theorem 3).** An exact, sampling-free formula for the expected ordered measure of independent route durations with given marginals (we use moment-matched gamma marginals), as an integral of a Poisson–binomial distribution against tail sums of the weights. It handles negative weights and costs $O(Gm^2)$.

- **Rank prices (Proposition 4).** The derivative of expected unfairness with respect to a location shift of a route's duration equals the expected weight of its rank. Prices of dispersion measures sum to zero, and a negative price is precisely the stochastic incentive to lengthen a route.

- **Risk-consistent tours (Proposition 6, Theorem 8).** Even $\mathbb{E}\max$ can prefer a non-shortest tour. We define consistency through the driver's own mean–standard deviation and prove that the consistent tour is a supported Pareto tour of (length, quadratic length), found exactly by dichotomic parametric Held–Karp.

- **Algorithm and evidence.** SORTIE, a bilevel local search whose moves are screened by rank prices, is compared with the deterministic fair plan, SAA with up to 5000 scenarios, a risk-adjusted deterministic baseline and ablations, in seventeen experiments: on the instances used by the deterministic literature and on a second family of twelve CVRPLIB instances, with exact enumeration on small instances, budget, fleet-size, noise-level and risk-attitude sweeps, four noise models including correlated noise, and up to 100 customers.

2 Literature review

SORTIE sits at the intersection of four literatures: workload equity in deterministic vehicle routing, inequity-averse optimisation with ordered measures, fairness under uncertainty, and routing with stochastic travel times. We review each in turn, stating for the closest works what they model, how they solve it and what they leave open, and close with the tools we borrow from probability and multicriteria optimisation. Table 1 summarises the positioning.

2.1 Workload equity in deterministic routing

The systematic study of balanced routing starts with Matl et al. [2018], who compare the equity functions used in the literature (min-max, range, mean absolute deviation, Gini-type measures and others) on the same instances. Their central observation is that the measures differ in *monotonicity*: a measure that is not monotone in the individual workloads can be improved by making a short route longer, so an optimal "fair" plan may contain routes that are needlessly inefficient. Matl et al. [2019a] show that the choice of the workload resource (distance, duration, load or number of customers) changes the balanced plan and the cost of equity, and Matl et al. [2019b] approximate the cost–balance Pareto front with a box-splitting scheme around a single-objective heuristic. In all three works the workload is a deterministic quantity attached to a route.

Exact methods have followed two directions. The min-max (makespan) objective has a long tradition [Applegate et al., 2002], and Bertazzi et al. [2015] quantify how much longer the longest route can become when total cost is minimised instead. For balance objectives, Halvorsen-Weare and Savelsbergh [2016] study bi-objective general routing with several balance criteria, Bektaş and Letchford [2020] propose ℓ_p norms of the workload vector as a tunable family between efficiency and min-max fairness, and Agius et al. [2026] develop exact algorithms for routing with route-cost equity *constraints*. van Rossum et al. [2024, 2026a] design branching rules for range and order-based objectives that make the range tractable in branch-and-price, and van Rossum et al. [2026b] add cutting planes that forbid arc sequences dominated by a shorter tour of the same customers, which removes exactly the inefficiency identified by Matl et al. [2018]. Related models balance tours in the multiple travelling salesman problem with a bound on the Gini coefficient of tour lengths [Bhadoriya et al., 2026], trade egalitarian against utilitarian welfare in cooperative fleets [López Sánchez et al., 2024], weigh the cost of driver satisfaction [Bruinink et al., 2025], equalise *cumulative* workload across periods by assigning daily routes to drivers [Nekooghadirli et al., 2026], and allocate workload among heterogeneous drivers by Nash social welfare under a cost cap [Mahmoodian et al., 2025], and design equity-oriented workload and pay for crowdsourced drivers [Sobhanan et al., 2026].

The direct predecessor of this paper is Ljubić et al. [2026]. They write a whole family of fairness objectives (maximum, range, Gini mean difference, absolute deviation, variance) as ordered measures of the vector of route lengths, borrowing the ordered-median machinery of location theory, and resolve the monotonicity problem of Matl et al. [2018] with a *bilevel* formulation: the upper level chooses the partition of the customers, and each lower level must route its customers on a shortest tour. They solve the resulting X-CVRP-TSP exactly on instances sliced from CVRPLIB X-n641-k35, which we reuse. Their model, like every work in this subsection, is deterministic; their conclusion lists uncertain demands as future work and does not discuss travel-time uncertainty. SORTIE keeps their bilevel structure and their ordered measures but lets route durations be random. Three things change. The objective becomes an expectation of an ordered measure of random variables, which no deterministic formulation represents (Section 4). Monotonicity no longer guarantees that the optimal tours are shortest ones (Proposition 6), so route consistency must be imposed for every measure, not only for the non-monotone ones. And "shortest" itself needs a definition when the duration of a tour is a random variable; we give one through each driver's mean–standard-deviation criterion and show that it is computable (Theorem 8).

2.2 Ordered measures and inequity-averse optimisation

Ordered weighted averaging was introduced by Yager [1988] as an aggregation that weights outcomes by their rank rather than by their identity; with nondecreasing weights it is convex and contains the maximum, the range and the Gini mean difference as special cases. In location theory the same idea became the ordered median problem [Nickel and Puerto, 2005], with linear programming representations of sorted sums through top-k sums [Ogryczak and Tamir, 2003, Ogryczak and Śliwiński, 2003] and a theory of equitable location built on inequality measures [Ogryczak, 2000]. Recent work extends the machinery to equity within facilities [Blanco et al., 2024], to Benders decomposition for the discrete ordered median problem [Ljubić et al., 2024], to quadratic ordered measures such as the variance [Blanco et al., 2026], and to the ordered median travelling salesman problem, in which the edges of a single tour are weighted by rank [Ljubić et al., 2026]. The broader inequity-averse literature is surveyed by Karsu and Morton [2015], and Tsang and Shehadeh [2026] characterise a class of convex fairness measures and show how stable optimal solutions are with respect to the choice of measure. In all of these works the ordered measure is applied to a deterministic vector. Our Theorem 3 gives the expectation of an ordered measure of independent random variables in closed integral form, and Proposition 4 its derivative with respect to each component; we are not aware of an analogue in the ordered-median literature.

2.3 Fairness under uncertainty

Only a handful of papers combine routing, fairness and uncertainty, and none of them concerns the fairness of drivers' random workloads. Kinable et al. [2026] study a stochastic vehicle routing problem with partial deliveries in which demand is random and fairness is defined among *customers*, as a minimum expected fill rate with a Rawlsian criterion; the quantity to be shared is the service received, not the work done. Gao et al. [2026] combine sequential fair allocation with routing under demand uncertainty and define max-min objectives over locations, both ex post (per realisation) and in a forward sense; again the recipients of fairness are locations, and the measure is max-min rather than a general ordered measure. On the workforce side, Anh [2022] assign tasks to workers with uncertain service times so that the difference in total working time stays below a threshold with high probability, using a distributionally robust joint chance constraint approximated by worst-case CVaR; there is no routing. Lin et al. [2022] design two-echelon routes for meal delivery

with a workload-fairness criterion among workers and protect it against service-time uncertainty with robust constraints; the fairness measure is evaluated on the deterministic side of the robust counterpart, not as a distribution of realised workloads. Gig-economy and rideshare work treats fairness of assignment *probabilities* and of matching under stochastic arrivals [Singh et al., 2023, Nanda et al., 2020] and fair allocation of delivery tasks [Basik et al., 2021, Hosseini and Schierreich, 2025]; the object is an assignment, not the duration of a route.

The word *consistency* is used differently in the consistent vehicle routing problem, where it means that customers are visited by the same driver at similar times across days [Groër et al., 2009, Kovacs et al., 2015]. This notion has been extended to stochastic customers and demands with a two-stage model solved by sample average approximation and decomposition [Alvarez et al., 2024], to robust demand [Wu et al., 2025] and to driver equity with flexible consistency [Yu et al., 2024]. It is unrelated to the route consistency of Ljubić et al. [2026] and of this paper, which requires each route to be an optimal tour of its own customers.

The distinction between ex-ante and ex-post fairness that organises our model is well established outside routing. In welfare economics, Fleurbaey and Peragine [2012] contrast equality of prospects with equality of outcomes. Gutjahr [2021] formalises ex-ante and ex-post inequity aversion in stochastic decision processes and relates them to time consistency, and Mostajabdaveh et al. [2019] combine ex-ante and ex-post Gini-based inequity in stochastic shelter location, followed by Alem et al. [2022] for two-stage humanitarian location–allocation and by Mostajabdaveh et al. [2025] for a bi-objective relief routing problem with a Gini index of unsatisfied demand. Sinclair et al. [2023] characterise the envy–efficiency trade-off of sequential allocation under stochastic demand, Shehadeh [2023] reduce disparities in transport distance in stochastic facility location, Ye et al. [2024] define distributional fairness through Wasserstein distances, and Barak and Talgam-Cohen [2026] use exactly our two definitions (the measure of expected loads against the expected measure) in facility-location mechanism design. These works establish that the two notions differ and that the ex-post one is the relevant one for affected individuals; none of them concerns routing or ordered measures of random route durations. We bring the distinction to fair routing, quantify the gap between the two notions (Proposition 5) and show how to optimise the ex-post notion exactly in the objective.

2.4 Stochastic travel times in vehicle routing

Routing with random travel times has a rich literature, but its objectives are cost, feasibility or reliability rather than equity. Laporte et al. [1992] introduce a priori routes with chance-constrained and recourse models, Kenyon and Morton [2003] include a makespan-type objective on the expected or quantile completion time of the longest route, and Bakach et al. [2021] study makespan objectives with correlated travel times; these are the closest to our maximum measure, but they optimise a moment or quantile of the makespan, not an ordered measure of all routes, and they do not consider the dispersion of workloads. Distribution fitting and correlation are discussed by Gómez et al. [2016], stochastic and robust formulations with deadlines by Adulyasak and Jaillet [2016], a priori optimisation with recourse for time windows by Errico et al. [2016, 2018], the cost–service trade-off by Zhang et al. [2013], service-level guarantees by Ehmke et al. [2015], and CVaR-type risk in latency location-routing by Osorio-Mora et al. [2025]; the surveys of Oyola et al. [2018] and Gendreau et al. [2016] give the wider picture. Correlated travel times are handled exactly by Rostami et al. [2021] and approximately by Rajabi-Bahaabadi et al. [2019], Bomboi et al. [2021]; robust and distributionally robust variants include Han et al. [2014], Jaillet et al. [2016], Zhang et al. [2021] and Ghosal and Wiesemann [2020]. Sample average approximation [Verweij et al., 2003] is the generic alternative to our closed-form evaluation, and we compare against it directly (E15).

Table 1: Positioning of SORTIE among the closest works. Drivers: fairness among drivers' workloads; Ordered: general ordered measures; Random durations: route durations are random; Ex post: fairness of realised outcomes is optimised; Consistency: every route is an optimal tour of its customers. (○: partly, see text.)

Work	Routing	Drivers	Ordered	Random durations	Ex post	Consistency
Matl et al. [2018]	✓	✓	○			
van Rossum et al. [2026b]	✓	✓	○			✓
Ljubić et al. [2026]	✓	✓	✓			✓
Kenyon and Morton [2003], Bakach et al. [2021]	✓	○		✓	○	
Kinable et al. [2026]	✓					
Gao et al. [2026]	✓				✓	
Lin et al. [2022]	✓	✓		○		
Anh [2022]		✓		✓	○	
Mostajabdaveh et al. [2019], Gutjahr [2021]			○		✓	
SORTIE (this paper)	✓	✓	✓	✓	✓	✓

Mean–risk path choice [Sen et al., 2001, Nikolova and Stier-Moses, 2014] motivates our definition of a risk-consistent tour as one that minimises the driver's mean plus a multiple of the standard deviation. Conditional value-at-risk [Rockafellar and Uryasev, 2000, 2002], which we report for the realised range, and the price of fairness [Bertsimas et al., 2011, Mo and Walrand, 2000], which E9 traces for stochastic fairness, complete the background.

2.5 Order statistics and parametric optimisation

Our evaluator rests on classical results on order statistics of independent, non-identically distributed variables. Their joint distribution can be written with permanents [Vaughan and Venables, 1972, Bapat and Beg, 1989], which are expensive to evaluate [Balakrishnan, 2007, Glueck et al., 2008]; the alternative we use expresses the event "at most r variables lie below t" through the Poisson–binomial distribution of the number of successes in independent non-identical trials [Hoeffding, 1956], whose probability mass function is computed exactly by a short recursion [Fernandez and Williams, 2010, Hong, 2013, Biscarri et al., 2018]; see also David and Nagaraja [2003] and Sen [1970]. To our knowledge the Poisson–binomial representation has not been used to evaluate or differentiate ordered fairness measures in optimisation. For the lower level, supported efficient solutions of bicriteria problems are enumerated by dichotomic (weighted-sum) search [Aneja and Nair, 1979], see Ehrgott [2005], Ehrgott and Gandibleux [2000], Ralphs et al. [2006], and small travelling salesman problems are solved by the Held–Karp recursion [Held and Karp, 1962, Bellman, 1962].

3 Model

Let $G = (V, A)$ be a complete graph with depot 0, customers $V' = \{1, \dots, n\}$, symmetric costs c_{ij} satisfying the triangle inequality, demands q_i and m identical vehicles of capacity Q. A plan consists of a partition $S_1, \dots, S_m$ of V' into nonempty capacity-feasible sets and a tour r_k of $\{0\} \cup S_k$ for each vehicle.

Random workloads. The travel time of arc (i, j) is $\tau_{ij} = Y c_{ij} \varepsilon_{ij}$, where $Y = \exp(s_Y Z - s_Y^2/2)$, $Z \sim N(0, 1)$, is a fleet-wide day factor with $\mathbb{E}Y = 1$, and the arc disturbances ε_{ij} are independent of each other and of Y with mean 1 and variance v^2 (gamma in our experiments). The workload

of driver k is $\Phi_k = YX_k$ with

$$X_k = \sum_{(i,j)\in r_k} c_{ij}\varepsilon_{ij}, \qquad \mathbb{E}X_k = L_k = \sum_{(i,j)\in r_k} c_{ij}, \qquad \operatorname{Var} X_k = \sigma_k^2 = v^2 W_k, \quad W_k = \sum_{(i,j)\in r_k} c_{ij}^2.$$

L_k is the tour length and W_k its *quadratic length*. Given Y the workloads are independent; unconditionally they are positively correlated.

Ordered fairness. For $a \in \mathbb{R}^m$ with order statistics $a_{(1)} \le \cdots \le a_{(m)}$ and weights $\lambda \in \mathbb{R}^m$, $\ell_\lambda(a) = \sum_j \lambda_j a_{(j)}$. We use the maximum ($\lambda = e_m$), the range ($e_m - e_1$), the GMD ($\lambda_j = 2(2j-m-1)/m^2$) and the mean absolute deviation from the median (MAD; $\lambda = (-1,\dots,-1,0,1,\dots,1)/m$ for odd m), together with the empirical variance $\mathrm{Var}_{\mathrm{emp}}(a) = m^{-1}\sum_k (a_k - \bar{a})^2$. All four λ are nondecreasing.

Definition 1 (Three notions of fairness)**.** For a plan with random workloads Φ: *ex-ante* unfairness $\ell_\lambda(\mathbb{E}\Phi)$; *ex-post* unfairness $\mathbb{E}\ell_\lambda(\Phi)$; *risk of unfairness* $\mathrm{CVaR}_\beta\, \ell_\lambda(\Phi)$.

A deterministic fair model optimises the ex-ante notion. Drivers experience the ex-post one. SORTIE optimises ex-post unfairness and reports the CVaR out of sample.

4 Theory

Proposition 2 (Common shocks)**.** *For every λ,*

$$\mathbb{E}\ell_\lambda(\Phi) = \mathbb{E}\ell_\lambda(X) \qquad \textit{and} \qquad \mathbb{E}\operatorname{Var}_{\mathrm{emp}}(\Phi) = e^{s_Y^2}\,\mathbb{E}\operatorname{Var}_{\mathrm{emp}}(X).$$

Proof. Sorting commutes with multiplication by $y > 0$, so $\ell_\lambda(yX) = y\ell_\lambda(X)$; take expectations using independence of Y and X and $\mathbb{E}Y = 1$. The empirical variance is homogeneous of degree two and $\mathbb{E}Y^2 = e^{s_Y^2}$. □

Fleet-wide shocks therefore drop out of the planning objective for ordered measures, which reduces evaluation to the conditionally independent vector X.

Theorem 3 (Poisson–binomial order-statistic integral)**.** *Let $X_1,\dots,X_m \ge 0$ be independent with CDFs F_k. For $t \ge 0$ let $N(t) = \#\{k : X_k \le t\}$, a Poisson–binomial count with success probabilities $F_k(t)$, and $\pi_r(t) = \mathbb{P}(N(t) = r)$. With tail sums $\Lambda_r = \sum_{j=r+1}^m \lambda_j$,*

$$\mathbb{E}\ell_\lambda(X) = \int_0^\infty \sum_{r=0}^{m-1} \Lambda_r\, \pi_r(t)\, dt. \tag{1}$$

Proof. $X_{(j)} > t$ if and only if fewer than j of the X_k are at most t, so $\mathbb{E}X_{(j)} = \int_0^\infty \mathbb{P}(N(t) \le j-1)\, dt$. Hence $\mathbb{E}\ell_\lambda(X) = \int_0^\infty \sum_j \lambda_j \sum_{r\le j-1} \pi_r(t)\, dt = \int_0^\infty \sum_r \pi_r(t) \sum_{j\ge r+1} \lambda_j\, dt$, where Tonelli's theorem is applied to the positive and negative parts of λ separately. □

The vector $\pi(t)$ is the coefficient vector of $\prod_k (1 - F_k(t) + F_k(t)z)$, obtained by m convolution steps. Below the smallest lower quantile the integrand equals Λ_0, above the largest upper quantile it vanishes ($\Lambda_m := 0$), and in between we use Gauss–Legendre panels whose breakpoints are quantiles of *every* marginal, so that a short, low-variance route is resolved even when the other routes are long. We use moment-matched gamma marginals $X_k \sim \Gamma(L_k^2/\sigma_k^2, \sigma_k^2/L_k)$. The identity behind (1) is classical for order statistics of non-identical variables [Bapat and Beg, 1989, Balakrishnan, 2007]; our contribution is to make it an exact planning objective for ordered fairness, where the tail sums Λ_r absorb the negative weights of dispersion measures, and to differentiate it.

Proposition 4 (Rank prices)**.** *Let R_k be the rank of X_k and define $\varpi_k = \sum_j \lambda_j \mathbb{P}(R_k = j)$, where $\mathbb{P}(R_k = j) = \int f_k(t)\,\mathbb{P}(N_{-k}(t) = j-1)\,dt$ and N_{-k} counts the other drivers. Then $\frac{d}{d\delta}\mathbb{E}\ell_\lambda(X + \delta e_k)\big|_{\delta=0} = \varpi_k$ and $\sum_k \varpi_k = \sum_j \lambda_j$.*

Proof. ℓ_λ is piecewise linear with $\partial \ell_\lambda / \partial a_k = \lambda_{R_k(a)}$ off the null set of ties and is $\|\lambda\|_1$-Lipschitz, so differentiation and expectation commute by dominated convergence. The ranks form a permutation, so $\sum_k \mathbb{P}(R_k = j) = 1$. □

For the range, GMD and MAD the prices sum to zero: burdened drivers carry positive prices and light drivers negative ones. A negative price is exactly the incentive to lengthen a route. Deterministically, the price of the lightest route under the range is -1, and a detour of length δ reduces the range by δ; stochastically, the incentive is spread over all drivers that might be the lightest on a given day. SORTIE uses the prices, together with $\psi_k = \partial \mathbb{E}\ell_\lambda / \partial \sigma_k$, to predict the effect of a move.

Proposition 5 (Fairness illusion)**.** *If λ is nondecreasing then ℓ_λ is convex and*

$$0 \le \mathbb{E}\ell_\lambda(X) - \ell_\lambda(L) \le \|\lambda\|_1\, \mathbb{E}\|X - L\|_\infty \le \|\lambda\|_1\, v\Big(\sum_k W_k\Big)^{1/2}.$$

Moreover $\mathbb{E}\,\mathrm{Var}_{\mathrm{emp}}(X) = \mathrm{Var}_{\mathrm{emp}}(L) + \frac{m-1}{m^2}\sum_k \sigma_k^2$, and for Gaussian marginals $\mathbb{E}\,\mathrm{GMD}(X) = m^{-2}\sum_{i\neq j}\left[s_{ij}\sqrt{2/\pi}\,e^{-d_{ij}^2/(2s_{ij}^2)} + d_{ij}(1 - 2\Phi(-d_{ij}/s_{ij}))\right]$ with $d_{ij} = L_i - L_j$ and $s_{ij}^2 = \sigma_i^2 + \sigma_j^2$.

Proof. For nondecreasing λ the rearrangement inequality gives $\ell_\lambda(a) = \max_\sigma \sum_j \lambda_j a_{\sigma(j)}$, a maximum of linear functions; Jensen's inequality gives the lower bound. Since $|a_{(j)} - b_{(j)}| \le \|a - b\|_\infty$ for every j, ℓ_λ is $\|\lambda\|_1$-Lipschitz in the sup norm, and $\mathbb{E}\max_k |Z_k| \le (\mathbb{E}\sum_k Z_k^2)^{1/2}$. The variance identity follows from $\mathbb{E}X_k^2 = L_k^2 + \sigma_k^2$ and $\mathbb{E}\bar{X}^2 = \bar{L}^2 + m^{-2}\sum_k \sigma_k^2$; the GMD formula is the mean of a folded normal. □

A deterministic fair plan therefore *overstates* the fairness drivers will experience. The overstatement grows with the arc noise and with the quadratic lengths: long arcs, not long routes, are what make a balanced plan unbalanced in practice.

Proposition 6 (Monotonicity no longer protects drivers)**.** *With deterministic workloads, every monotone measure ($\lambda \ge 0$) has an optimal plan made of TSP-optimal routes [Matl et al., 2018]. With random workloads this fails even for $\mathbb{E}\max$.*

Example 7. Two drivers; $X_1 \equiv 100$. Driver 2 can drive tour A ($L = 100$, $\sigma = 10$) or tour B ($L = 101$, $\sigma \approx 0$). With Gaussian X_2, $\mathbb{E}\max = 100 + 10\varphi(0) \approx 103.99$ under A and ≈ 101 under B, so the optimiser picks the longer tour B.

Route consistency must therefore be imposed for every measure, and it needs a risk attitude. We let each driver rank the tours of their customer set by

$$\rho_\kappa(r) = \frac{\mathbb{E}[\Phi(r) \mid Y] + \kappa\,\mathrm{SD}[\Phi(r) \mid Y]}{Y} = L(r) + \kappa v\sqrt{W(r)},$$

which ranks tours identically on every day. $\kappa = 0$ is TSP-optimality (ties broken by smaller W).

Theorem 8 (Risk-consistent tours are supported)**.** *For $\kappa \ge 0$, some minimiser of ρ_κ over the tours of a fixed customer set is a supported extreme tour: it minimises $L(r) + \theta W(r)$ for some $\theta \in [0, \infty)$, or minimises W lexicographically after L ($\theta = \infty$). All supported extreme tours are enumerated by dichotomic search, each step being one Held–Karp run with arc weights $c_{ij} + \theta c_{ij}^2$.*

Proof. $f(L, W) = L + \kappa v\sqrt{W}$ is concave and nondecreasing on $\mathbb{R}^2_+$. A concave function bounded below on a pointed polyhedron attains its minimum at an extreme point; since f is nondecreasing, the minimum of f over the finite set $P = \{(L(r), W(r))\}$ equals its minimum over $\mathrm{conv}(P) + \mathbb{R}^2_+$ and is attained at an extreme point of that set. These extreme points are exactly the unique minimisers of $w_1 L + w_2 W$ with $(w_1, w_2) \geq 0$, $(w_1, w_2) \neq 0$ (for $w_1 = 0$ with ties broken by smaller L), and dichotomic search enumerates all of them [Aneja and Nair, 1979]. Tours in the relative interior of a hull edge are dominated by the edge's end points and need not be enumerated. With heterogeneous noise (ε_{ij} with variance $v^2 H_{ij}^2$) the same holds with $W = \sum (H_{ij} c_{ij})^2$ and arc weights $c_{ij} + \theta H_{ij}^2 c_{ij}^2$. □

The SORTIE problem.

$$
\begin{aligned}
\min\ & \mathbb{E}\, \ell_\lambda\big(X(r_1, \ldots, r_m)\big) \\
\text{s.t.}\ & (S_1, \ldots, S_m) \text{ capacity-feasible partition},\ S_k \neq \emptyset, \qquad \textstyle\sum_k L(r_k) \leq B \cdot C^*, \\
& r_k \in \arg\min\{\rho_\kappa(r) : r \text{ a tour of } \{0\} \cup S_k\} \quad \forall k,
\end{aligned}
$$

where C^* is the minimum expected total length and $B \geq 1$ the budget factor. With $v = 0$ and $\kappa = 0$ this is the X-CVRP-TSP of Ljubić et al. [2026].

5 The SORTIE algorithm

The SORTIE problem combines three difficulties: a capacitated partitioning problem in the upper level, a travelling salesman problem (or, for $\kappa > 0$, a bi-criteria tour problem) for every route in the lower level, and an objective that is an expectation over order statistics of random variables. Exact methods for the deterministic special case [Ljubić et al., 2026, van Rossum et al., 2026b] report instances with up to about 25 customers [van Rossum et al., 2026b]; the stochastic objective is non-linear in the route statistics and is not representable in their formulations. We therefore propose a bilevel local search that keeps the lower level exact wherever it is applied to a plan the search accepts, and uses the theory of Section 4 in two places: PBOI evaluates the objective without sampling, and the rank prices predict the effect of a move before it is evaluated. This section describes the representation and lower level (§5.1), the evaluator (§5.2), the first-order screen (§5.3), the exact confirmation that keeps every accepted plan bilevel feasible (§5.4), the ruin-and-recreate perturbation and acceptance rule (§5.5), and the parameters and complexity (§5.6). Algorithms 1 and 2 give the pseudocode.

5.1 Representation and lower level

A solution is a list of m customer sequences, one per vehicle. The upper level is the partition; the sequence of each route is not a decision of the upper level but the output of the lower-level solver applied to the route's customer set S. For $\kappa = 0$ the lower level returns a shortest tour of $\{0\} \cup S$, with ties broken by smaller quadratic length. For $|S| \leq 15$ it is solved exactly by the Held–Karp recursion [Held and Karp, 1962] compiled with numba, in $O(2^{|S|}|S|^2)$ time (about 5 ms at 15 customers); for larger sets we use 2-opt and Or-opt (segments of one to three customers, both orientations) from eight starts: a warm start built from the route's previous sequence by cheapest insertion of new customers, the nearest-neighbour tour, and random permutations. For $\kappa > 0$ the lower level minimises $\rho_\kappa = L + \kappa v\sqrt{W}$ by the dichotomic search of Theorem 8: starting from the tour minimising L (ties by W) and the tour minimising W (ties by L), it repeatedly solves the weighted problem with arc weights $c_{ij} + \theta d_{ij}^2$, where θ is the slope between two known supported

Algorithm 1 SORTIE: main loop

Require: instance, measure λ, noise level v, risk weight κ, budget $B\,C^*$, time limit

$P \leftarrow$ cost-efficient plan; re-solve every route with the exact lower level; repair cost if $\sum_k L_k > B\,C^*$
$P^* \leftarrow P$; $T_0 \leftarrow 0.02\,|\mathbb{E}\ell_\lambda(X(P))|$; *stuck*$\leftarrow$NONE
while time remains **do**
  compute rank prices ϖ and dispersion prices ψ of P (Prop. 4)
  if *stuck*$\neq P$ **then**
    $P' \leftarrow$ImproveStep(P, ϖ, ψ) ▷ Algorithm 2
    if $P' \neq$NONE **then** $P \leftarrow P'$; update P^*; **continue**
    else *stuck*$\leftarrow P$
    end if
  end if
  $P'' \leftarrow$RuinRecreate(P, ϖ, ψ); repair cost; confirm changed routes exactly
  accept P'' with probability $\min\{1, \exp(-(F(P'') - F(P))/T)\}$, $T = T_0\,(10^{-3})^{\text{elapsed/limit}}$
  update P^*
end while
return P^*

tours and $d_{ij} = H_{ij}c_{ij}$ (so $d = c$ under homogeneous noise), until no new supported tour appears; every weighted problem is solved like the $\kappa = 0$ case. Lower-level results are memoised by customer set, so a set that reappears during the search is never solved twice.

5.2 Evaluation

Given the tour lengths L_k and quadratic lengths W_k, every route has mean L_k and standard deviation $\sigma_k = v\sqrt{W_k}$, and its duration is modelled by a gamma marginal with these two moments (normal marginals are an option; E16 shows that the choice hardly matters). The objective $F(P) = \mathbb{E}\ell_\lambda(X)$ is evaluated by the integral of Theorem 3. The integrand is smooth between the quantiles of the marginals and changes quickly near them, so we integrate by six-point Gauss–Legendre rules on the panels delimited by the quantiles of every marginal at 13 levels from 10^{-12} to $1-10^{-12}$; with $m = 5$ this gives about 360 nodes. At each node the Poisson–binomial probability mass function of the number of routes finished by time t is computed by the standard recursion over routes in $O(m^2)$, so one evaluation costs $O(Gm^2)$ for G nodes, about 0.4 ms. The rank prices of Proposition 4 reuse the same nodes, with the mass function of the other $m-1$ routes, in $O(Gm^3)$; the dispersion prices $\psi_k = \partial F/\partial\sigma_k$ are obtained by forward differences with step $0.05\,\sigma_k$, i.e. m further evaluations. For the expected empirical variance all quantities are in closed form (Proposition 5). Prices are computed once per accepted plan.

5.3 First-order screening

The neighbourhood consists of all *relocate* moves (customer i from route a to route b) and all *swap* moves (customers $i \in a$ and $j \in b$ exchanged), about $n(m-1) + n^2(m-1)/(2m)$ moves in total. Evaluating every move exactly would require two lower-level solves and one PBOI evaluation per move. Instead, for every route we precompute, in vectorised form, the change in L and W caused by removing each of its customers (bridging its neighbours), by inserting any customer at its cheapest position, and by replacing each of its customers by any other customer at the cheapest position in the shortened route. These insertion deltas approximate the change of the route statistics, and the

Algorithm 2 ImproveStep(P, ϖ, ψ): screen, evaluate, confirm

for every relocate $(i : a \to b)$ and swap $(i \in a,\ j \in b)$ **do**
  compute insertion and removal deltas $\Delta L_a, \Delta L_b, \Delta W_a, \Delta W_b$
  if capacity holds and $\sum_k L_k + \Delta L_a + \Delta L_b \le (1 + \epsilon_B)\, B\, C^*$ **then**
    $\hat{\Delta} \leftarrow \varpi_a \Delta L_a + \varpi_b \Delta L_b + \psi_a \Delta\sigma_a + \psi_b \Delta\sigma_b$ ▷ eq. (2)
  end if
end for
$\mathcal{K} \leftarrow$ the K moves with the smallest $\hat{\Delta}$
for each move in $\mathcal{K}$ **do**
  re-solve routes a, b with *screening* tours; evaluate F by PBOI
end for
for each improving candidate, best first **do**
  re-solve routes a, b with the *exact* lower level; re-evaluate F
  if F still improves and $\sum_k L_k \le B\, C^*$ **then return** the confirmed plan
  end if
end for
return NONE

change of the objective is predicted to first order by the prices:

$$\hat{\Delta} = \sum_{k \in \{a,b\}} \Big(\varpi_k\, \Delta L_k + \psi_k\, v\big(\sqrt{W_k + \Delta W_k} - \sqrt{W_k}\big) \Big). \tag{2}$$

Moves that violate capacity, or whose predicted total length exceeds the budget by more than $\epsilon_B = 2\%$, are discarded; the remaining moves are ranked by $\hat{\Delta}$ and the best $K = 8$ are evaluated. By Proposition 4, ϖ_k is the exact derivative of F with respect to a shift of route k, so (2) is the linearisation of F in the route statistics, with the insertion deltas standing in for the re-optimised tours. A negative ϖ_k (possible for the range, the GMD and the MAD) predicts a gain from *lengthening* route k, which is exactly the incentive the lower level neutralises; the screen therefore proposes such moves, and the exact evaluation then reveals whether they survive the re-optimisation of the tour. E14 shows that the prediction ranks moves with a Spearman correlation of about 0.9 with their exact effect. For the deterministic baseline the same screen is used with the deterministic rank prices (λ at the rank of each route, $\psi = 0$), for SAA with the empirical rank frequencies over the scenarios, and for Det-σ with the deterministic prices of $L + \sigma$, so all methods share the search and differ only in the objective.

5.4 Exact confirmation

The K candidates are evaluated with *screening tours*: a single 2-opt/Or-opt descent from the warm start (for $\kappa > 0$ inside the same dichotomic search, capped at 24 tours), without Held–Karp. Improving candidates are then confirmed in order of their screened value: both changed routes are re-solved by the exact lower level, the objective is re-evaluated, and the move is applied only if it still improves and respects the budget. Plans produced by the perturbation are confirmed route by route in the same way.

Proposition 9 (Bilevel feasibility of accepted plans)**.** *Every plan that Algorithm 1 accepts as its current or best plan is capacity feasible, satisfies $\sum_k L_k \le B\, C^*$, and routes every customer set on the tour returned by the exact lower level.*

Proof. The start plan is re-solved by the exact lower level and repaired by cost descent, whose steps are themselves confirmed moves. A plan becomes current only through ImproveStep, which returns a plan only after re-solving the two changed routes exactly and re-checking the budget, or through the perturbation, whose output is accepted only after every route that differs from the current plan has been re-solved exactly and the budget re-checked. All other routes are unchanged and were exact before. Capacity is checked for every move and every insertion. □

The screening tour is never shorter than the exact one, so a move that improves only with exact tours can be missed; this affects the quality of the search, not feasibility. E14 shows that 28 of 30 final SORTIE plans admit no improving relocate or swap move at all, i.e. the search ends in genuine local optima of the neighbourhood. Together with a cap on the ruin size, lazy confirmation raises the iteration rate of SORTIE-0 at $n = 100$ from about 250 to about 1700 per minute, and at $n = 75$ from about 170 to about 1800.

5.5 Perturbation and acceptance

When no screened move improves the current plan, SORTIE applies a ruin-and-recreate perturbation. The number of removed customers q is drawn uniformly from $\{2, \dots, \max(3, \min(12, \lceil n/4 \rceil))\}$. With probability $1/2$ the removal pool is *price-weighted*: routes are ordered by sampling without replacement with probabilities proportional to $|\varpi_k|$, and their customers are added to the pool until it holds at least q; otherwise the pool is all customers. From the pool, with probability $1/2$ the q customers closest to a random seed customer are removed (related removal), otherwise q random customers; no route is emptied. The removed customers are reinserted by *price-scored regret-2 insertion*: inserting customer i at its cheapest position in route b is scored by

$$s_b(i) = \varpi_b \, \Delta L_b + \psi_b \, \Delta\sigma_b + 10^{-3} \, \Delta L_b,$$

the customer with the largest difference between its best and second-best capacity-feasible score is inserted first, and the scores of the modified route are updated. The small cost term breaks ties in favour of shorter insertions. If the result exceeds the budget it is repaired by cost descent: ImproveStep with unit prices and zero dispersion prices, so that $\hat{\Delta}$ is the change in total length, for at most 200 steps; if the budget cannot be met the perturbation is discarded. The repaired plan is confirmed exactly (§5.4) and accepted by simulated annealing with temperature $T = T_0(10^{-3})^{\tau}$, where τ is the elapsed fraction of the time limit. The initial temperature $T_0 = 0.02 \, |\mathbb{E}\ell_\lambda(X(P_0))|$ is computed from the PBOI value of the start plan for every method, so that Det, SAA and SORTIE accept uphill moves of the same absolute size.

5.6 Parameters and complexity

Table 2 lists all parameters; none was tuned per instance. E14 shows that the number of evaluated candidates K between 1 and 32 does not change the median result. One improvement step costs the precomputation of the deltas, $O(m \, n_k^2 N)$ for routes of n_k customers and $N = n + 1$ nodes, which is vectorised; the ranking of all moves, $O(n^2)$; K screening re-solves of two routes, each a local-search descent; K PBOI evaluations, $O(KGm^2)$; and at most a few exact re-solves. The prices cost $O(Gm^3) + m$ PBOI evaluations per accepted plan. On the main grid an iteration takes 7 to 9 ms (median), and the search runs 2300 to 3100 iterations in 15 to 25 s; at $n = 100$ it runs about 3500 iterations in 120 s (E7).

Table 2: Parameters of SORTIE (identical for all methods and instances).

Component	Parameter	Value
Lower level	exact Held–Karp up to	15 customers
	local-search starts beyond that	8
	screening tours	1 descent, $\leq$24 tours if $\kappa > 0$
Evaluator	Gauss–Legendre points per panel / quantile levels	6 / 13
	dispersion-price step	$0.05\,\sigma_k$
Screen	candidates evaluated per step K	8
	budget slack in the screen ϵ_B	2%
Ruin	removed customers q	2 to $\max(3, \min(12, \lceil n/4 \rceil))$
	price-weighted pool / related removal	probability 1/2 each
Recreate	cost tie-breaker in $s_b(i)$	10^{-3}
Repair	maximum cost-descent steps	200
Annealing	initial temperature / final ratio	$0.02\,\lvert F(P_0)\rvert$ / 10^{-3}

6 Computational study

6.1 Set-up

Instances. We follow the slicing rule of the deterministic fair routing literature [Ljubić et al., 2026]. Block b of size n consists of the depot and the n consecutive customers $bn+1, \ldots, (b+1)n$ of CVRPLIB X-n641-k35 [Uchoa et al., 2017], with ceiled Euclidean distances, $m = 5$ vehicles and capacity $Q = \lfloor \sum_i q_i/4 \rfloor$, so one vehicle's worth of slack is available. The main grid uses $n \in \{15, 20, 25, 30, 35\}$ and blocks $b = 0, \ldots, 4$ (25 instances); the scale test uses $n \in \{50, 75, 100\}$ and blocks 0–5; the exactness test uses ten instances with $n = 10$, $m = 3$ and $Q = \lfloor \sum_i q_i/2.4 \rfloor$.

Uncertainty. Unless stated otherwise $v = 0.3$ (gamma arc noise) and $s_Y = 0.15$. In the congestion experiment (E8) the arc coefficient of variation is vH_{ij}, where $H_{ij} \in [1/3, 2]$ is largest for arcs near a randomly drawn congestion centre, so noise is no longer proportional to length.

Budget. C^* is the cost of the best of three restarts of the cost-minimising search; the total length of every fair plan must satisfy $\sum_k L_k \leq B\,C^*$ with $B \in \{1.05, 1.10\}$.

Methods. All methods share the search of Algorithm 1, the start plan, the time limit (15 s for $n \leq 25$, 25 s for $n \geq 30$, 120 s for $n \geq 50$) and the random seed. They differ only in the objective and in the lower level. *Det* minimises the deterministic ordered measure $\ell_\lambda(L)$ of the tour lengths, i.e. the model of Ljubić et al. [2026] solved heuristically. *SAA* minimises the sample average of $\ell_\lambda(X)$ over 200 fixed arc scenarios. *SORTIE-0* minimises the exact ex-post value by PBOI with TSP-optimal tours ($\kappa = 0$); *SORTIE-1* uses the risk-consistent lower level with $\kappa = 1$. In Det and SAA the moves are screened by the deterministic and the sample objective respectively.

Evaluation. Every plan is evaluated out of sample by arc-level Monte Carlo with 20 000 scenarios (10 000 for $n \geq 50$), including the day factor, with common random numbers across methods. The reported *ex-post value* is the Monte Carlo estimate of $\mathbb{E}\ell_\lambda(\Phi)$. Paired comparisons use the Wilcoxon signed-rank test. All tables are generated from the raw results by the released scripts.

Table 3: E1: PBOI against arc-level Monte Carlo (50 000 samples) on 168 plans; z: difference over its Monte Carlo standard error.

Measure	Plans	Mean |err| (%)	Max |err| (%)	Share $\|z\| < 3$
Max	168	0.041	0.137	0.994
Range	168	0.062	0.249	1.000
GMD	168	0.057	0.241	1.000
MAD	168	0.064	0.272	0.988
Var	168	0.108	0.440	0.994

Table 4: E3: what each method believes its unfairness is, relative to the true ex-post value (Monte Carlo without the day factor, which is what each model represents), mean signed error over 50 runs per cell. Last column: largest absolute error of the PBOI prediction for any method's plan.

	Believed vs true (mean signed %)				PBOI
Measure	Det	SAA	SORTIE-0	SORTIE-1	max |err| %
Max	-7.97	-0.71	-0.01	-0.01	0.20
Range	-17.56	-0.99	+0.03	+0.02	0.70
GMD	-14.68	-0.91	+0.04	+0.05	0.61
MAD	-16.17	-0.88	+0.05	+0.07	0.68
Var	-27.23	-1.33	+0.00	+0.02	0.98

6.2 Accuracy and cost of the evaluator (E1, E2)

Table 3 compares PBOI with 50 000-sample arc-level Monte Carlo on 168 plans: the cost-efficient plans and random feasible plans whose routes are sequenced by the lower level, for $n \in \{15, 25, 35, 50\}$. The mean absolute relative error is below 0.11% for all five measures and at most 0.44% in the worst case, and 99–100% of the differences lie within three Monte Carlo standard errors. The mean z-scores are not exactly zero (up to 0.34 for the MAD): PBOI carries a systematic bias of a few hundredths of a percent, which we attribute to the moment-matched gamma marginal of each route (a sum of arc gammas with different scales is not exactly gamma). PBOI takes 0.44 ms per plan for $m = 5$, against 0.28 s for 20 000-scenario Monte Carlo. A 200-scenario SAA is about seven times faster than PBOI but is a noisy estimate. The advantage of PBOI is therefore that it is exact and noise-free at a cost compatible with thousands of evaluations per second, not raw speed.

On the ten $n = 10$ instances we enumerated all capacity- and budget-feasible partitions (4 to 145 per instance) with exact Held–Karp tours. The cost search found the optimal C^* on all ten. For both range and GMD, SORTIE-0 found the plan with the smallest exact ex-post value on nine of ten instances (mean gap 0.39%, maximum 3.9%). Det found the deterministic optimum on nine of ten; on the tenth instance, which has only 20 budget-feasible partitions, it stopped at a plan with twice the optimal deterministic range.

6.3 The fairness illusion and the value of ex-post optimisation (E3)

Table 4 quantifies Proposition 5. A deterministic fair plan believes its unfairness is the ex-ante value, which understates the true ex-post value by 8% for the maximum, 15–18% for the range, GMD and MAD, and 27% for the variance. The in-sample SAA value is optimistic by about 1%, which is the optimiser's curse on 200 scenarios. The PBOI value believed by SORTIE has a mean signed error below 0.1%, a mean absolute error of 0.07–0.23% and a worst case of 1%.

Table 5 reports the out-of-sample ex-post value relative to Det. The signed-rank test favours

Table 5: E3: improvement of the out-of-sample ex-post value over Det (mean %, positive is better; in brackets the number of instances on which the method is strictly better). *: Wilcoxon signed-rank $p < 0.05$ in favour of the method; the test is on ranks and can disagree in sign with the mean when one instance dominates (Var, $B = 1.10$, SORTIE-1).

Measure	B	SAA	SORTIE-0	SORTIE-1
GMD	1.05	-0.40 (6/25)	-0.09 (8/25)	-0.03 (9/25)
GMD	1.10	-0.32 (9/25)	+0.94* (14/25)	+0.34* (17/25)
MAD	1.05	+0.86* (11/25)	+1.10* (13/25)	+1.24* (12/25)
MAD	1.10	+0.70 (13/25)	+0.96* (14/25)	+1.53* (18/25)
Max	1.05	+0.71* (14/25)	+0.61* (15/25)	+0.77* (15/25)
Max	1.10	+1.70* (23/25)	+2.08* (24/25)	+1.93* (24/25)
Range	1.05	+0.12 (13/25)	+0.64* (15/25)	+1.56* (15/25)
Range	1.10	-0.21 (13/25)	+1.07* (17/25)	+0.39* (15/25)
Var	1.05	+0.48 (7/25)	+0.50* (9/25)	+0.48* (13/25)
Var	1.10	-2.25 (6/25)	+1.09* (12/25)	-0.49* (16/25)

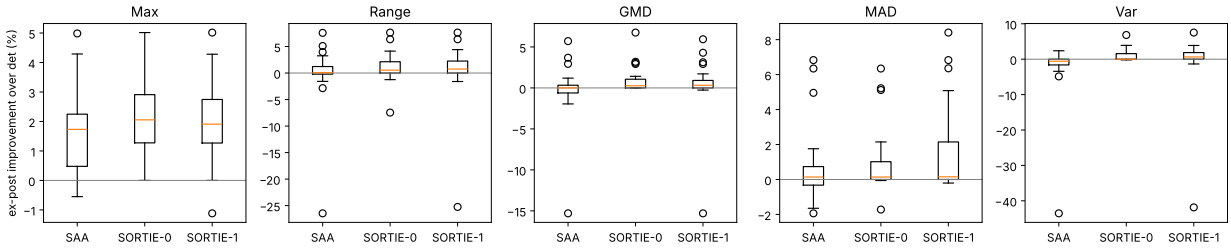


Figure 1: E3, $B = 1.10$: per-instance improvement of the ex-post value over Det.

SORTIE-0 and SORTIE-1 over Det in 9 of the 10 cells each. Pooled over all 250 runs, SORTIE-0 improves on Det by 0.89% on average (median 0.23%; better in 141 runs, tied in 96, worse in 13; $p < 10^{-18}$) and SORTIE-1 by 0.77% (better in 154, worse in 31). SAA improves by only 0.14% (median zero; better in 115, worse in 74) and is favoured significantly in 3 cells. Head to head, SORTIE-0 beats SAA in 145 runs and loses in 15, and SORTIE-1 beats it in 147 and loses in 27 ($p < 10^{-17}$). The largest gains are for the maximum at $B = 1.10$ (2.1%, 24 of 25 instances), where the extra budget lets the planner trade a slightly longer but less variable longest route.

The improvements are small, about one percent on average, and this is itself a finding. With homogeneous noise ($\sigma_k^2 = v^2 W_k$) the standard deviation of a route grows with its length, so the plans that balance lengths also nearly balance risks, and a deterministic fair plan is close to ex-post optimal even though its *assessment* of its own fairness is badly wrong. For context, the cost-optimal plan has an ex-post range of 2087 against about 1340 for the fair plans, at a cost 6.5% higher: fairness itself is worth optimising; the stochastic refinement adds about one percent on top. For range-optimised plans the $\mathrm{CVaR}_{0.9}$ of the realised range, which none of the methods optimises, is 2054 for Det, 2059 for SAA, 2037 for SORTIE-0 and 2032 for SORTIE-1; both SORTIE variants improve it at both budgets, SAA at neither.

An illustrative instance. Figure 2 shows the cost-optimal plan and the range-optimised Det and SORTIE-0 plans of X641-n30-b0 at $B = 1.10$, an instance whose SORTIE-0 gain (+0.47%) is close to the median of the 25 instances (+0.54%). The cost plan leaves one driver with a route of

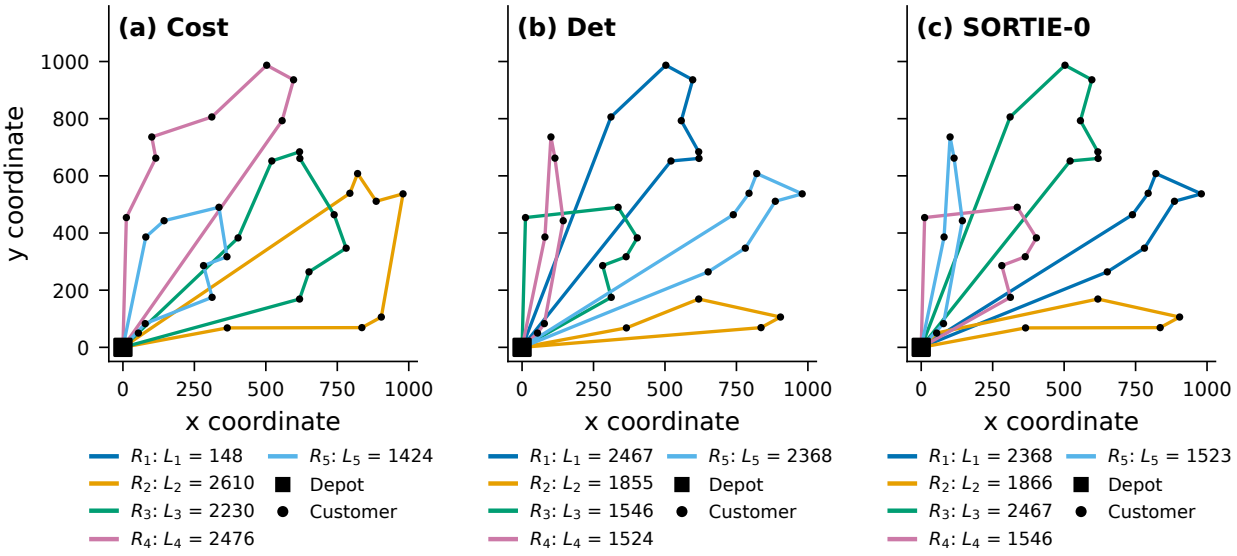


Figure 2: X641-n30-b0, $B = 1.10$: cost-optimal plan and the range-optimised Det and SORTIE-0 plans. L_k: tour lengths.

length 148 and another with 2610. Both fair plans balance the lengths to a deterministic range of about 940, and they differ only in how two route pairs are split. Figure 3 shows what the drivers experience. The realised workloads are wide and right-skewed, and the expected realised range is about 1225 for both fair plans: 30% above the deterministic range the Det planner reports. Figure 4 gives the distribution of the realised range of the two plans; in one day out of ten it exceeds 1750. The worst instance for SORTIE-0 in this cell is X641-n35-b0, where Det is 7.5% better (Figure 21 in Appendix A); this is consistent with the search noise documented in E7b rather than a systematic effect, since on the same instance SORTIE-0 is better or tied in eight of the other nine cells.

How tight is the illusion bound? Figure 5 plots the gap between the true ex-post value and the ex-ante value of every Det plan against the bound of Proposition 5. No plan violates the bound. The bound is conservative: the realised gap is on average 33% of the bound for the maximum, 18% for the range and 11–12% for the GMD and MAD, because it charges every route its full standard deviation while only the routes near the extreme ranks matter. Its value is that it is computable from the quadratic lengths alone and grows like $v\sqrt{\sum_k W_k}$, which is the scaling observed in E6.

6.4 Route consistency (E4)

E4 removes the lower level: the planner may sequence each route freely (*free*) instead of using the TSP-optimal tour (*lower*). For the maximum, free sequencing changes almost nothing, as monotonicity predicts in the deterministic case. For the range and the GMD the free planner lengthens the short routes: 23 of 25 Det plans and 24 of 25 SORTIE plans for the range contain a route that is not a shortest tour of its customers; the excess length is 95 (Det) and 113 (SORTIE) distance units on average over the runs that have one. This lowers the ex-post value by 1–3% for Det on both measures and for SORTIE on the range (Table 6); for SORTIE on the GMD it is 0.9% worse, because the larger space slows the search. Wherever it helps, the reduction comes from making some drivers drive detours. This is exactly the behaviour signalled by negative rank

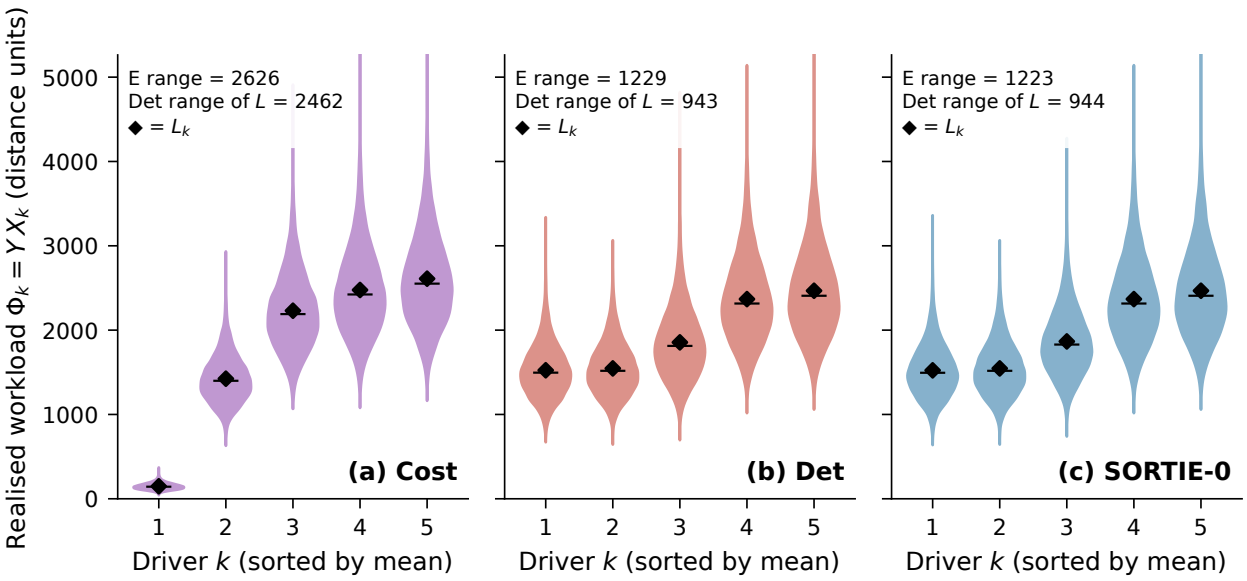


Figure 3: Simulated realised workloads $\Phi_k = YX_k$ of the three plans of Figure 2 (20 000 days); diamonds mark the tour lengths L_k that a deterministic model sees.

prices and excluded by the bilevel constraint. The deterministic belief error persists with free tours (-24% for the range); the PBOI belief remains exact.

6.5 Rank-price screening (E5)

Replacing the price screen by $K = 8$ random feasible moves, with everything else unchanged, gives slightly more iterations (3200 against 2700) but worse plans: price screening lowers the ex-post value by 5.6% for the range and 8.5% for the GMD on average (medians 0% and 5.4%; better on 12 and 17 of 25 instances, worse on 3 and 2, ties otherwise; $p = 0.005$ and $p < 10^{-3}$). For the range the mean is driven by a few instances on which random screening stalls far from a good plan; for the GMD the advantage is broad. The rank prices are thus an effective first-order guide for the search, and not only an interpretive device.

6.6 Noise level (E6)

Table 7 varies v for $n = 25$. The illusion grows monotonically with v, as the bound of Proposition 5 suggests: at $v = 0.5$ a deterministic plan understates its range by 43%. The gain of SORTIE over Det also grows with v for the range, from -0.5% at $v = 0.1$ (where the illusion is small and the differences are search noise) to 2.1% at $v = 0.5$ (better on all five instances); for the GMD it is 1.1%, 0.7% and 1.8%.

6.7 Larger instances (E7, E7b)

E7 runs $n \in \{50, 75, 100\}$ on six blocks each, with a 120 s limit, for the range, GMD and maximum (18 instance–measure pairs per size). With the screening lower level the search runs about 9000 iterations per run at $n = 50$ and 3500 at $n = 100$ for SORTIE-0 (Det: 15 000 and 4400). PBOI

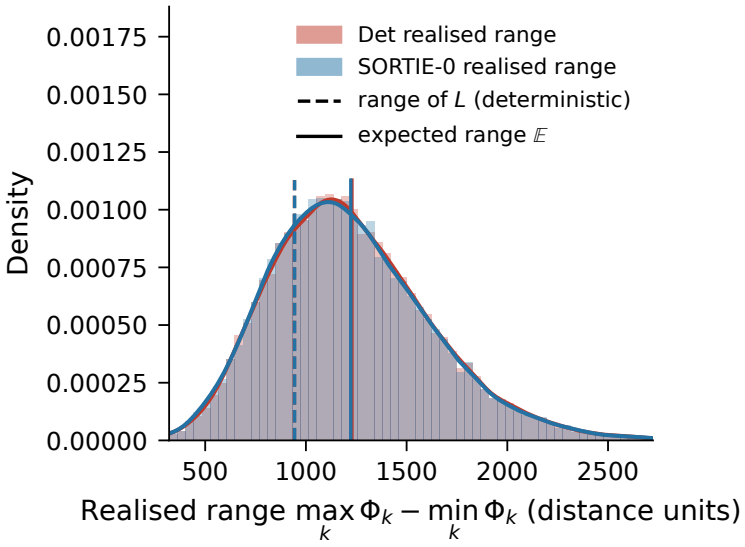


Figure 4: Distribution of the realised range of the Det and SORTIE-0 plans of Figure 2; dashed: deterministic range of the tour lengths, solid: expected realised range.

remains accurate (mean absolute error 0.15%, at most 0.9%) and the deterministic belief error is −17% on average, as on the small instances.

Table 8 pools the 18 pairs per size. SORTIE-0 improves on Det by about 2% at every size and is better on 12–13 of 18 pairs; the improvement is significant at $n = 50$ ($p = 0.035$) but not at $n = 75$ or $n = 100$ ($p = 0.17$). SORTIE-1 is significant at $n = 50$ but not beyond: its lower level runs a dichotomic search per route and completes only about 850 iterations at $n = 100$, and 56% of its runs there (39% at $n = 75$) still improved by more than 0.5% in the second half of the time limit. Single runs at $n \geq 75$ are thus dominated by search noise, especially for the range and the GMD, whose value depends on a few routes.

E7b separates the method from the search noise at $n = 100$: the six instances and three measures are solved by Det, SAA and SORTIE-0 with three solver seeds each and a 240 s limit, and each method is represented by the mean ex-post value over its seeds (Table 9). Averaged over seeds, SORTIE-0 improves on Det by 4.2% (median 2.5%, better on 13 of 18 pairs, $p = 0.021$); SAA improves by 3.2% ($p = 0.10$). The difference between SORTIE-0 and SAA (1.1%, 12 of 18) is not significant ($p = 0.25$), and no single measure is significant on six instances. The seed-to-seed variation is itself informative: for the range, the ex-post value of the Det plan varies by 11% across seeds, that of SORTIE-0 by 3%. The deterministic range is flat over many plans whose realised fairness differs, and the search picks among them arbitrarily; the ex-post objective distinguishes them. Even at 240 s, 19% of the SORTIE-0 runs improved their in-model value by more than 0.5% in the second half of the run, so the absolute values at $n = 100$ are not fully converged.

Figure 8 shows the anytime behaviour behind these numbers, and Figure 9 the seed-to-seed spread at $n = 100$.

6.8 Heterogeneous noise: congestion (E8)

E3 suggests that the stochastic refinement matters little when a route's variance is proportional to its length. E8 breaks that proportionality: arcs near a congestion centre have up to twice the base coefficient of variation and arcs far from it a third. We rerun the main grid ($B = 1.10$; range, GMD

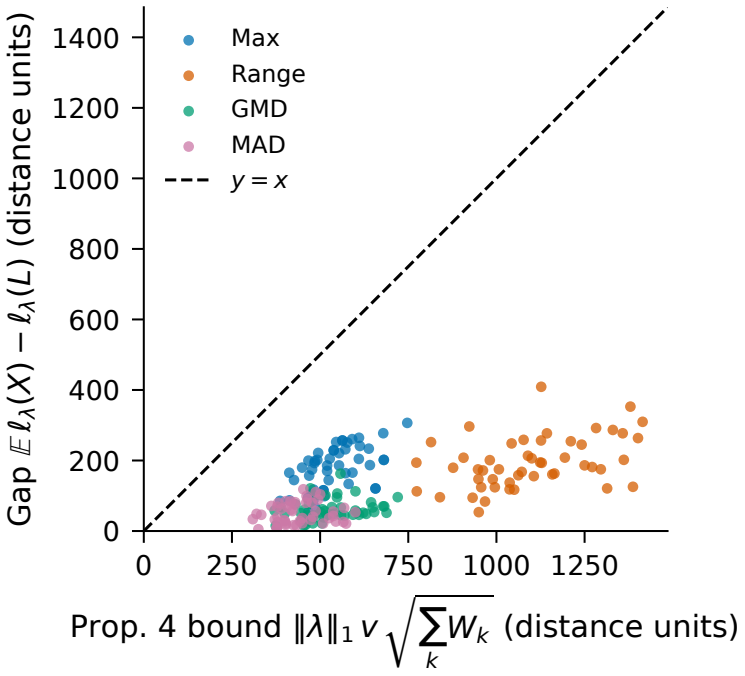


Figure 5: Fairness illusion of the Det plans of E3 (true ex-post minus ex-ante value) against the bound $\|\lambda\|_1 v(\sum_k W_k)^{1/2}$ of Proposition 5.

and max) on the congested versions of the same 25 instances, starting from the same cost plans.

The illusion becomes somewhat larger: a deterministic plan understates its range by 26%, its GMD by 22% and its maximum by 10%, against 23%, 19% and 9% on the same cells under homogeneous noise. The PBOI belief has a mean signed error below 0.2% (mean absolute error at most 0.26%). The gain of optimising ex-post fairness, however, grows only slightly (Table 10). Pooled over 75 runs, SORTIE-0 improves on Det by 1.51% (median 1.24%; better in 53, worse in 3; $p < 10^{-8}$) and SORTIE-1 by 1.66% (better in 57, worse in 10), against 1.36% for SORTIE-0 on the same cells without congestion. By measure the change is 0.94% $\to$ 1.40% for the GMD, 2.08% $\to$ 2.15% for the maximum and 1.07% $\to$ 0.96% for the range. SAA gains 1.28%, but with a median of only 0.47%, and SORTIE-0 beats it head to head in 49 of 75 runs (worse in 6; $p < 10^{-6}$). The $\mathrm{CVaR}_{0.9}$ of the range is 1919 for Det, 1905 for SAA, 1912 for SORTIE-0 and 1907 for SORTIE-1. Heterogeneous noise thus makes a deterministic plan's self-assessment much worse, but in our instances it changes the best plan only a little more than homogeneous noise does: the congested arcs are concentrated in one part of the network, and the assignment that balances lengths still roughly balances risk there.

Figure 10 shows the congestion multiplier on the illustrative instance, and Figure 11 pairs every instance and measure with and without congestion: the points scatter around the diagonal, with the GMD above it and the range on both sides.

6.9 The budget frontier (E9)

E9 repeats the comparison of Det and SORTIE-0 on the 10 instances with $n \in \{25, 35\}$ for budget factors B from 1.02 to 1.30 (Table 11, Figure 12). Two effects stand out. First, the gain of SORTIE-0 grows with the budget: from about 1% for $B \le 1.10$ to 3.2% at $B = 1.15$ and 6.8% at $B = 1.30$, where SORTIE-0 is better on all 30 instance–measure pairs. With little slack every

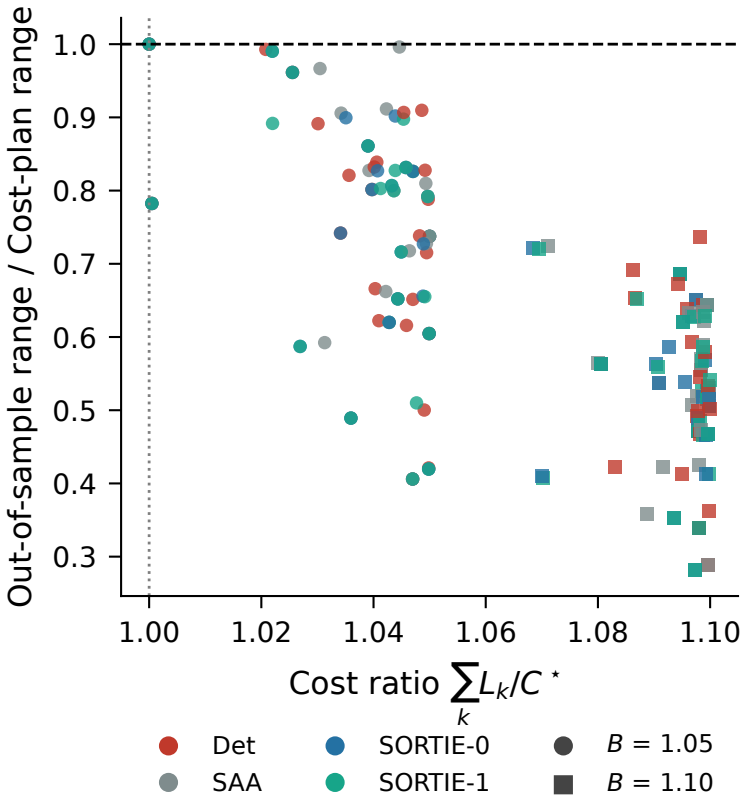


Figure 6: E3, range-optimised plans: total length relative to C^* against the expected realised range relative to that of the cost-optimal plan.

fair plan is close to the cost-optimal one and there is little room to choose; with more slack there are many plans whose deterministic range is about equal but whose realised range differs, and only the ex-post objective can tell them apart. Second, for $B \geq 1.15$ SORTIE-0 uses *less* of the budget (below that the total lengths agree within 0.3 percentage points). At $B = 1.30$ the Det plans for the maximum have a total length 20.5% above C^* and an expected realised maximum 5.2% below that of the cost plan; the SORTIE-0 plans use 13.5% and achieve 8.7%. For the maximum the deterministic objective even deteriorates from $B = 1.20$ to $B = 1.30$: the extra length it buys goes into routes whose deterministic length is irrelevant to the maximum but whose variance raises the realised one. At every budget level the ex-post objective reaches lower realised unfairness, and for generous budgets it does so with less total length.

The same cells at $B = 1.10$ appear in E3, and the per-cell means differ (for example -0.7% for the range in E9 against $+0.9\%$ in E3 on the same ten instances). The runs are identical except for wall-clock timing: the search stops at a time limit, so the iteration at which it stops, and hence the plan, can differ between two executions on a shared machine. This is another manifestation of the search noise analysed in E7b, and the reason we draw conclusions only from pooled comparisons.

6.10 Fleet size (E10)

E10 varies the number of vehicles on the $n = 35$ instances, with capacity $\lfloor \sum_i q_i/(m-1) \rfloor$ so that one vehicle's worth of slack remains (Table 12, Figure 13). The illusion grows with m: a deterministic plan understates its range by 10% with three vehicles, 24% with five and 27% with seven. More

Table 6: E4: free tours against lower-level tours, $B = 1.10$, 25 instances per row. [a]Runs with at least one route that is not a shortest tour of its customers. [b]Mean number of such routes. Excess: total length above the shortest tours of the same customer sets. Belief err: believed minus true ex-post value. Ex-post: out-of-sample ex-post value.

Measure	Method	Runs[a]	Routes[b]	Excess	Belief err (%)	Ex-post
Max	Det-lower	0/25	0.00	0.0	-8.94	2221.0
Max	Det-free	0/25	0.00	0.0	-9.01	2232.4
Max	SORTIE-lower	0/25	0.00	0.0	-0.00	2174.8
Max	SORTIE-free	7/25	0.36	2.7	+0.00	2177.1
Range	Det-lower	0/25	0.00	0.0	-23.11	1099.7
Range	Det-free	23/25	1.00	87.8	-24.19	1088.5
Range	SORTIE-lower	0/25	0.00	0.0	+0.04	1098.4
Range	SORTIE-free	24/25	1.64	108.4	+0.06	1072.6
GMD	Det-lower	0/25	0.00	0.0	-19.46	437.7
GMD	Det-free	21/25	1.12	90.0	-20.02	425.9
GMD	SORTIE-lower	0/25	0.00	0.0	+0.07	434.5
GMD	SORTIE-free	23/25	1.52	82.4	+0.10	438.6

Table 7: E6: arc noise v, $n = 25$, $B = 1.10$, five instances per row.

v	Measure	Det belief err (%)	SORTIE-0 vs Det
0.1	GMD	-6.4	+1.05 (2/5)
0.1	Range	-8.7	-0.54 (1/5)
0.3	GMD	-22.0	+0.65 (2/5)
0.3	Range	-28.7	+0.96 (3/5)
0.5	GMD	-35.6	+1.84 (4/5)
0.5	Range	-42.7	+2.14 (5/5)

routes compete for the extreme ranks, so more of the realised unfairness comes from which route happens to be longest on the day, which a deterministic model cannot see. The gain of SORTIE-0 is small with three and five vehicles (0.5% and 0.6%; with $m = 5$ not significant on these 15 pairs, $p = 0.06$) and 3.2% with seven (better on 13 of 15 pairs, $p = 0.001$). With $m = 3$ the GMD is proportional to the range (4/9 of it), so the two rows coincide.

6.11 Common shocks in practice (E11)

E11 re-evaluates all 500 E3 plans with $B = 1.10$ under day-factor dispersions $s_Y \in \{0, 0.15, 0.3, 0.5\}$, using the same arc draws (Table 13, Figure 14). As Proposition 2 predicts, the expected ordered measures are unchanged: the ratio of the value with and without the day factor is 1.000 up to Monte Carlo error (at most 1.0007 on average, at $s_Y = 0.5$). The expected empirical variance is multiplied by 1.023, 1.095 and 1.285, against $e^{s_Y^2} = 1.023$, 1.094 and 1.284. The tail is a different matter: the $\mathrm{CVaR}_{0.9}$ of the realised range grows by 7%, 25% and 58%. A fleet-wide shock is irrelevant for the choice of a plan by expected ordered fairness, but not for the risk of a very unfair day.

6.12 Robustness to the noise model (E12)

PBOI assumes independent route durations with given means and variances, and SORTIE was tuned to gamma arc noise. E12 re-evaluates the $B = 1.10$ plans of E3 (500 plans) under four arc-

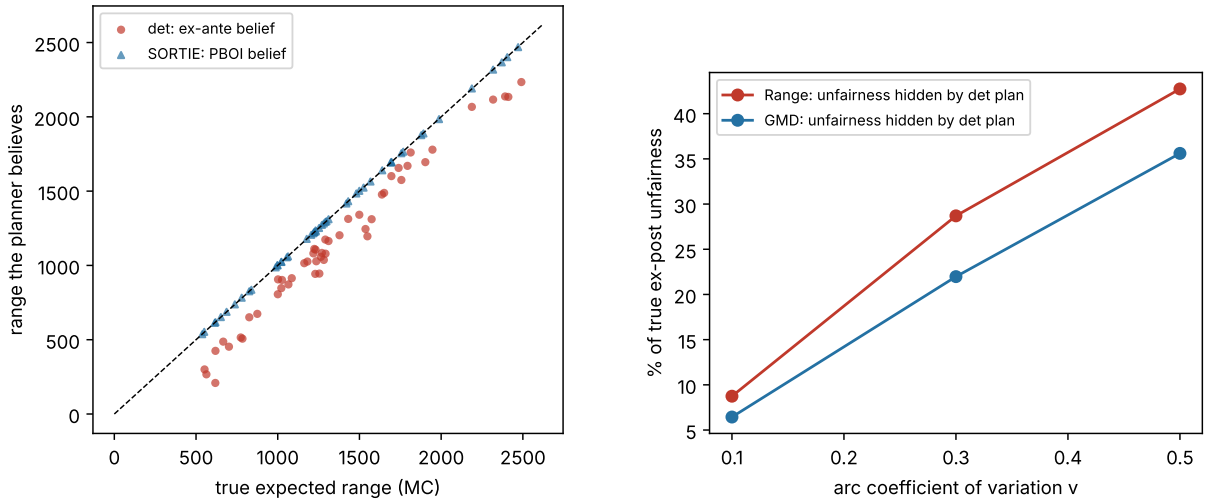


Figure 7: Left: range believed by Det (ex-ante) and by SORTIE (PBOI) against the true expected range, E3. Right: share of the true ex-post unfairness that a deterministic plan does not see, against v (E6).

Table 8: E7: improvement of the ex-post value over Det, $B = 1.10$, 120 s, pooled over six instances and three measures (range, GMD, max); format as Table 5.

n	SAA	SORTIE-0	SORTIE-1
50	+1.56 (12/18)	+2.04* (13/18)	+1.20* (15/18)
75	+1.96 (11/18)	+2.04 (13/18)	-0.29 (11/18)
100	+1.52 (10/18)	+2.21 (12/18)	-0.04 (8/18)

noise families, and those of E8 (300 plans) under the first two (the other two cannot reach the low per-arc variability of the uncongested arcs of E8) with the same mean and coefficient of variation per arc: gamma (the planning model), lognormal (heavier right tail), *incident* (with probability 0.02 an arc takes 2.5 times as long, the rest of the variability being gamma), and *regional* (a multiplicative factor shared by all arcs whose tail lies in the same quadrant of the plane, which makes routes in the same region positively correlated and so violates the independence that PBOI assumes) (Table 14, Figure 15).

The accuracy of PBOI degrades gracefully. Under lognormal noise its mean absolute error rises from 0.16% to 0.29%; under incidents to 1.4%, with a bias of +1.4% (it overstates unfairness) and a worst case of 9.5%; under regional correlation to 1.9%, with a worst case of 20% for the variance measure, the one most affected by correlation. What matters for planning is that the ranking of plans survives: under every family SORTIE-0 remains significantly better than Det, by 1.0% to 1.2% on the E3 plans and 1.5% on the E8 plans (gamma and lognormal), against 1.2% and 1.5% under the planning model.

6.13 Risk attitude of the drivers (E13)

E13 solves the $n \in \{25, 35\}$ instances with the risk-consistent lower level for $\kappa \in \{0, 0.5, 1, 2, 4\}$ (Table 15, Figure 16). Under homogeneous noise the risk-consistent tour rarely differs from the shortest one: even at $\kappa = 4$ only 14–16% of the routes change, and the extra length is about 13

Table 9: E7b, $n = 100$, $B = 1.10$, 240 s, three solver seeds: improvement of the seed-averaged ex-post value over Det (six instances per measure; format as Table 5) and coefficient of variation of the ex-post value across seeds, averaged over instances.

	vs Det, mean of 3 seeds		Seed CV (%)		
Measure	SAA	SORTIE-0	Det	SAA	SORTIE-0
Range	+6.45 (4/6)	+8.68 (5/6)	11.3	4.7	3.0
GMD	+2.49 (3/6)	+3.18 (4/6)	6.4	2.7	4.5
Max	+0.78 (4/6)	+0.86 (4/6)	1.4	0.8	1.4
All	+3.24 (11/18)	+4.24* (13/18)	SORTIE-0 vs SAA: +1.09 (12/18)		

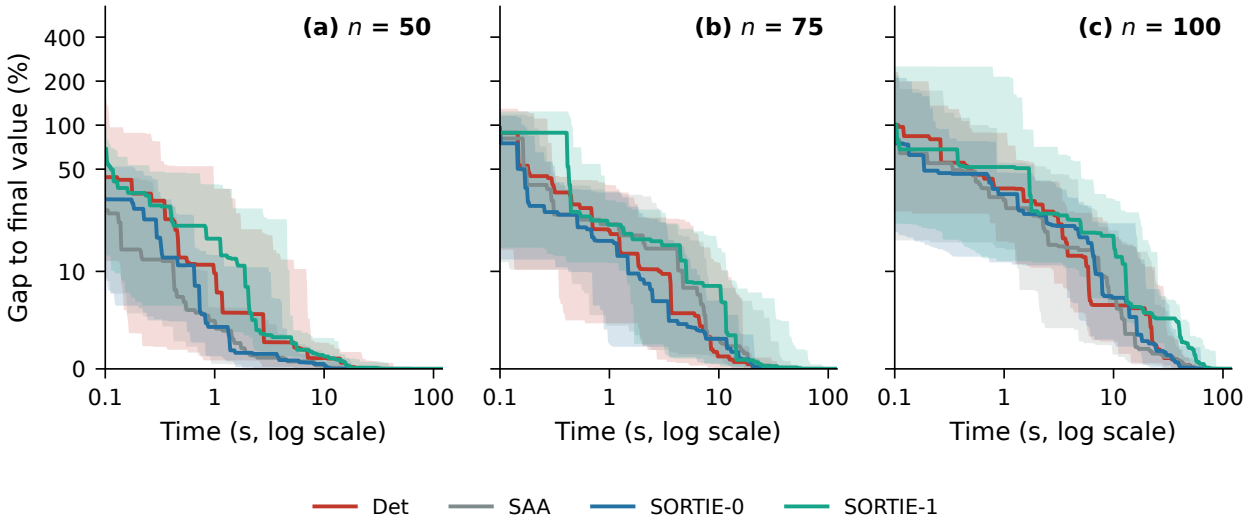


Figure 8: E7: best in-model value relative to the final value over the run (median and interquartile range over the 18 runs per method and size).

distance units per plan, under 0.2% of the total. The ex-post value changes by less than the search noise (median ratio to $\kappa = 0$ equal to one, interquartile range within $\pm 1\%$). The reason is that, for a fixed customer set, a tour's length and its quadratic length are nearly comonotone: the shortest tour is also nearly the least variable one. Proposition 6 shows that this need not be so, and with heterogeneous noise the two can separate, but on these instances the choice of κ is a matter of the drivers' preferences rather than of fairness.

6.14 Why the rank prices work (E14)

E14 checks the first-order model behind the screen directly. For the cost-optimal start plan and the final SORTIE-0 plan of the ten $n \in \{25, 35\}$ instances, we evaluated up to 400 relocate and swap moves both by the rank-price prediction and exactly (exact lower level and PBOI); 5577 moves in total (Table 16, Figures 17 and 18). The prediction ranks the moves almost as the exact evaluation does: Spearman's ρ is 0.89–0.94 on the start plans and 0.78–0.89 on the final plans. On the start plans the exact-best improving move is among the 8 best predicted moves in 80% of the cases (random ordering: 7%) and always among the 21 best. On the final plans of SORTIE-0 there is no improving relocate or swap move in 28 of 30 cases: the search ends in genuine local optima of

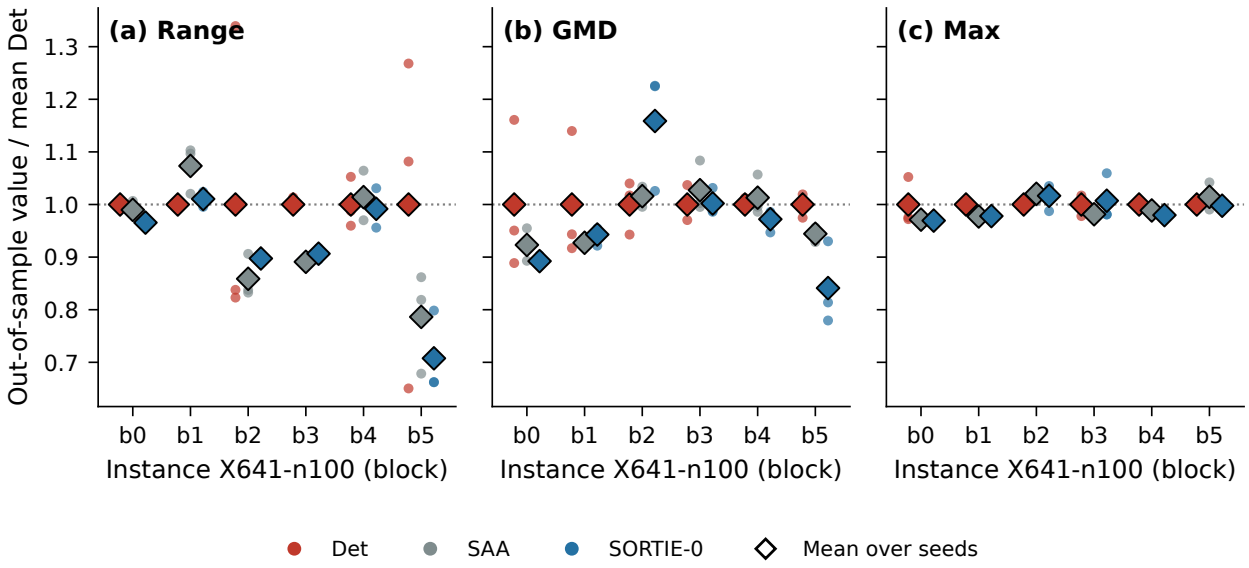


Figure 9: E7b, $n = 100$: ex-post value of each solver seed, relative to the mean Det value of the instance; larger markers: seed means.

Table 10: E8: congested arc noise, $B = 1.10$, 25 instances per row; improvement of the ex-post value over Det (format as Table 5).

Measure	SAA	SORTIE-0	SORTIE-1
GMD	+0.76 (11/25)	+1.40* (13/25)	+1.82* (15/25)
Max	+1.87* (23/25)	+2.15* (23/25)	+2.21* (24/25)
Range	+1.22 (13/25)	+0.96* (17/25)	+0.94* (18/25)

its neighbourhood. Consistently, the number of candidates K evaluated per step barely matters (Table 17): the median ex-post value is the same for K from 1 to 32.

6.15 Is a larger sample enough? (E15)

A natural objection is that SAA with 200 scenarios is simply too small. E15 reruns SAA with $S \in \{50, 200, 1000, 5000\}$ scenarios on the $n \in \{25, 35\}$ instances (Table 18, Figure 19). SORTIE-0 is better than every SAA variant on average: by 1.4% ($S = 50$), 1.6% ($S = 200$), 1.1% ($S = 1000$) and 0.9% ($S = 5000$), significantly for $S \leq 1000$. The gap shrinks as the sample grows, as it should, since SAA converges to the objective that PBOI evaluates exactly; with 5000 scenarios SAA is still worse on 9 pairs and better on 2 (the rest tied), but the difference is no longer significant at the 5% level ($p = 0.0505$). The sample average is cached per route, so the number of iterations drops only from about 5300 to 4200, and the remaining gap is the sampling error of the objective, not a lack of search time.

6.16 Simpler stochastic baselines (E16)

Two simpler ways of taking uncertainty into account are tested on the full E3 grid at $B = 1.10$ (Table 19). *Det-σ* applies the deterministic ordered measure to risk-adjusted lengths $L_k + \sigma_k$, the

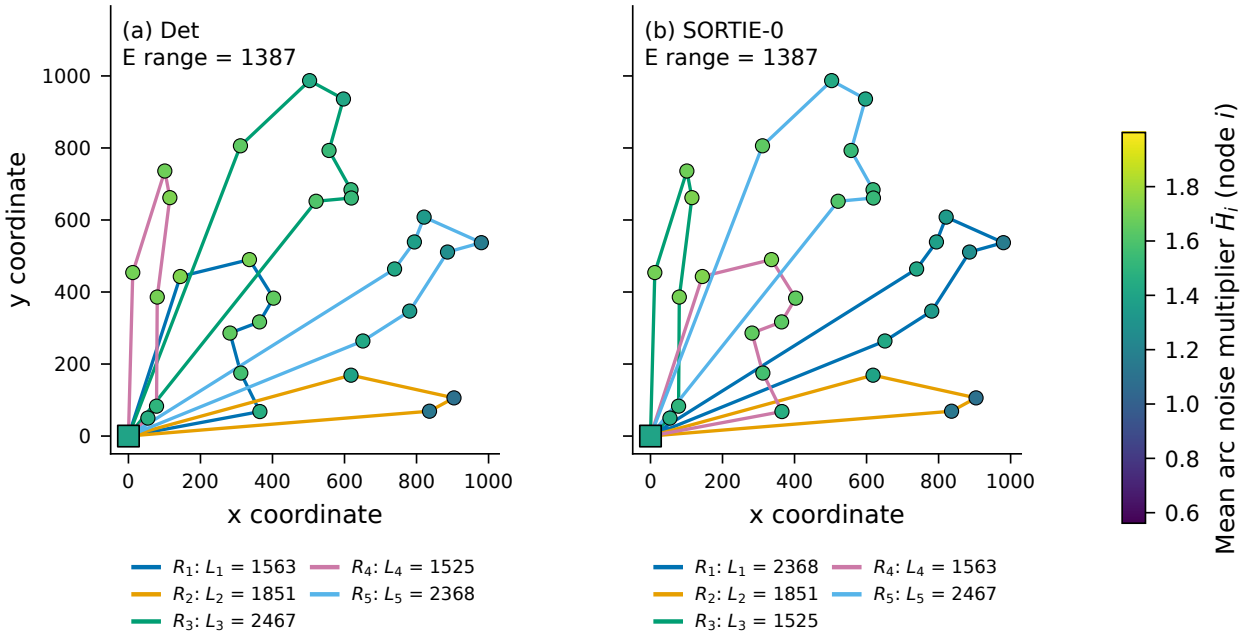


Figure 10: X641-n30-b0 with congestion: customers coloured by the mean noise multiplier H of their arcs, with the range-optimised Det and SORTIE-0 plans of E8.

fairness analogue of the robust-length heuristics common in stochastic routing. It is *worse* than Det (-0.8%, better on 37 and worse on 61 of 125 pairs, $p = 0.02$), and SORTIE-0 beats it by 1.9% (97 of 125, $p < 10^{-14}$). Adding a standard deviation to every route treats each route's risk separately, whereas the expected ordered measure depends on how likely each route is to occupy each rank, which is what the Poisson–binomial integral and the rank prices capture. *PBOI-N* replaces the gamma marginals by normal ones; it performs like SORTIE-0 (no significant difference, $p = 0.65$), so the method is insensitive to the marginal family once the first two moments are right.

6.17 A second benchmark family (E17)

All instances so far are slices of one file, and slicing places the depot at the first node of each block. E17 tests twelve further CVRPLIB X instances of Uchoa et al. [2017] (X-n101-k25 to X-n153-k22), which were generated with different depot positions and customer distributions. From each we take the depot of the file and its first 50 customers, keep $m = 5$ and the capacity rule $Q = \lfloor \sum_i q_i/4 \rfloor$, and solve range, GMD and maximum at $B = 1.10$ with Det, SAA, SORTIE-0 and Det-σ, 120 s and two solver seeds; each method is represented by the mean ex-post value over its seeds (Tables 20 and 21, Figure 20).

The results are stronger than on the sliced instances. The illusion reaches 40% for the range and 35% for the GMD, against 23% and 19% on the sliced instances at the same budget; plausibly the file's own depot, which is often eccentric, makes the routes differ more in variance than a block depot does. SORTIE-0 improves on Det by 2.2% on average (median 1.7%; better on 31 and worse on 5 of 36 pairs, $p < 10^{-6}$), with a positive average on every one of the twelve instances (0.7% to 4.2%). SAA improves by 0.7%, and SORTIE-0 beats it on 33 of 36 pairs ($p < 10^{-7}$). Det-σ is again worse than Det on average (-0.8%; better on 13 and worse on 23 pairs, not significant here, $p = 0.15$). The conclusions of E3 and E16 thus carry over to instances with other depot positions

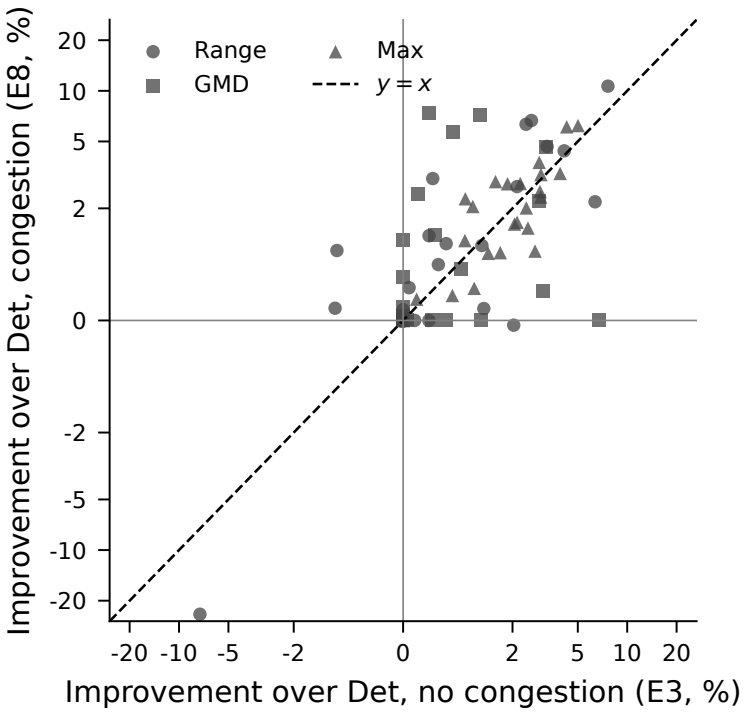


Figure 11: Improvement of SORTIE-0 over Det per instance and measure, without (E3) and with (E8) congestion, $B = 1.10$.

and customer distributions, with larger effects.

6.18 Managerial insights

The experiments support five statements a planner can act on.

(i) Report realised, not planned, fairness. A deterministic fair plan understates the unfairness its drivers will experience by 8–27% at a moderate noise level, by more with more vehicles (E10) and more noise (E6), and the understatement is not uniform across plans. PBOI computes the realised value in under a millisecond, so it can be reported for any plan, whatever method produced it.

(ii) Do not spend the budget on deterministic balance. With generous budgets a deterministic fair model spends the slack to equalise route lengths, which does little for realised fairness and can make it worse (E9). The ex-post objective achieves more and, with generous budgets, uses less: at $B = 1.30$ its expected unfairness is 6.8% lower on average (7.6% for the range) and its total length 3.4 percentage points closer to C^* on average, mostly because of the maximum.

(iii) Risk adjustment is not a substitute. Adding a safety margin to each route and then balancing (Det-σ) is worse than doing nothing (E16). What matters is the probability that each route ends up the longest or the shortest, which is what the rank prices measure.

(iv) More scenarios are not a substitute either. Sample average approximation needs thousands of scenarios to approach the exact objective, and with 5000 it is still worse on average, though no longer significantly (E15). The exact integral takes 0.4 ms per plan, whereas the 200-scenario average takes 0.06 ms, so a 5000-scenario average would, extrapolating linearly, cost several times more than PBOI.

(v) Keep the shortest-tour requirement. Without it, a fairness objective buys balance by sending drivers on detours (E4), which is exactly the behaviour the bilevel formulation of Ljubić et al. [2026]

Table 11: E9: improvement of the ex-post value of SORTIE-0 over Det by budget factor, $n \in \{25, 35\}$, ten instances per measure; cost ratio: mean $\sum_k L_k/C^*$ of the SORTIE-0 plans (format as Table 5).

B	Cost ratio	Pooled	Range	GMD	Max
1.02	1.014	+1.43* (17/30)	+1.40 (4/10)	+2.07 (5/10)	+0.82* (8/10)
1.05	1.041	+0.73* (19/30)	+0.81 (5/10)	+0.66 (5/10)	+0.73* (9/10)
1.10	1.094	+0.67* (20/30)	-0.72 (7/10)	+0.30 (3/10)	+2.45* (10/10)
1.15	1.134	+3.20* (27/30)	+5.92* (8/10)	+1.31 (9/10)	+2.37* (10/10)
1.20	1.170	+2.46* (26/30)	+2.59 (9/10)	+2.34* (7/10)	+2.46* (10/10)
1.30	1.222	+6.79* (30/30)	+7.64* (10/10)	+9.11* (10/10)	+3.63* (10/10)

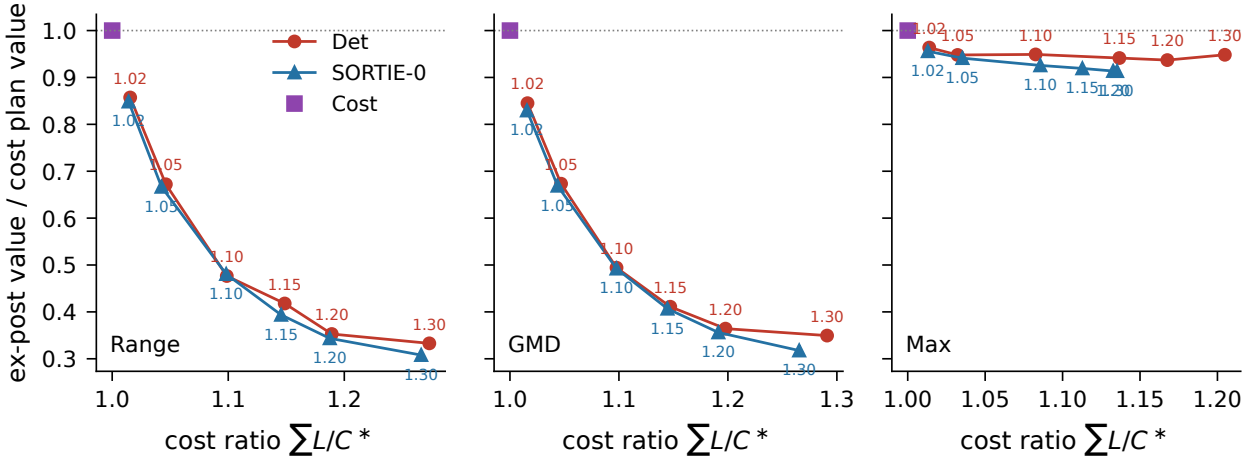


Figure 12: E9: price of fairness. Mean total length relative to C^* against the mean expected realised unfairness relative to the cost-optimal plan, for budget factors B from 1.02 to 1.30 (labels).

was designed to exclude; under uncertainty it must be imposed explicitly, and the risk-consistent variant changes few tours in practice (E13).

7 Conclusions

Fair routing under random travel times has to say whose fairness it measures. Drivers experience realised durations, so the relevant quantity is the expected unfairness of the realised day. A deterministic fair plan cannot see this quantity, and in our experiments it understated it by 8% (maximum) to 27% (variance) at $v = 0.3$, and by up to 43% at $v = 0.5$. A multiplicative common shock, by contrast, leaves the expected value of every ordered measure unchanged (Proposition 2), although it does inflate the variance measure and the tail of unfairness.

The Poisson–binomial order-statistic integral evaluates any ordered measure of independent route durations exactly, in under a millisecond for five vehicles, and its derivatives, the rank prices, are both interpretable and useful: they identify the routes a fairness objective would like to lengthen, and screening moves with them gave plans 6–9% better on average than random screening. Under uncertainty, route consistency no longer follows from monotonicity, so it has to be part of the model; the mean–standard-deviation lower level makes it computable by parametric Held–Karp. Without

Table 12: E10: fleet size, $n = 35$, five instances per measure, $B = 1.10$ (format as Table 5).

m	Measure	Det belief err. (%)	SAA vs Det	SORTIE-0 vs Det
3	Pooled	-8.2	+2.83* (7/15)	+0.48* (8/15)
3	Range	-9.9	+3.66 (2/5)	+0.14 (2/5)
3	GMD	-9.9	+3.66 (2/5)	+0.14 (2/5)
3	Max	-4.6	+1.17 (3/5)	+1.17 (4/5)
5	Pooled	-18.6	-1.62 (8/15)	+0.64 (9/15)
5	Range	-24.1	-3.11 (2/5)	+2.78 (3/5)
5	GMD	-23.2	-4.01 (1/5)	-3.58 (1/5)
5	Max	-8.5	+2.26 (5/5)	+2.71 (5/5)
7	Pooled	-20.3	+2.34* (10/15)	+3.16* (13/15)
7	Range	-26.7	+1.36 (3/5)	+4.58 (5/5)
7	GMD	-23.3	+3.34 (2/5)	+2.14 (3/5)
7	Max	-10.9	+2.31 (5/5)	+2.76 (5/5)

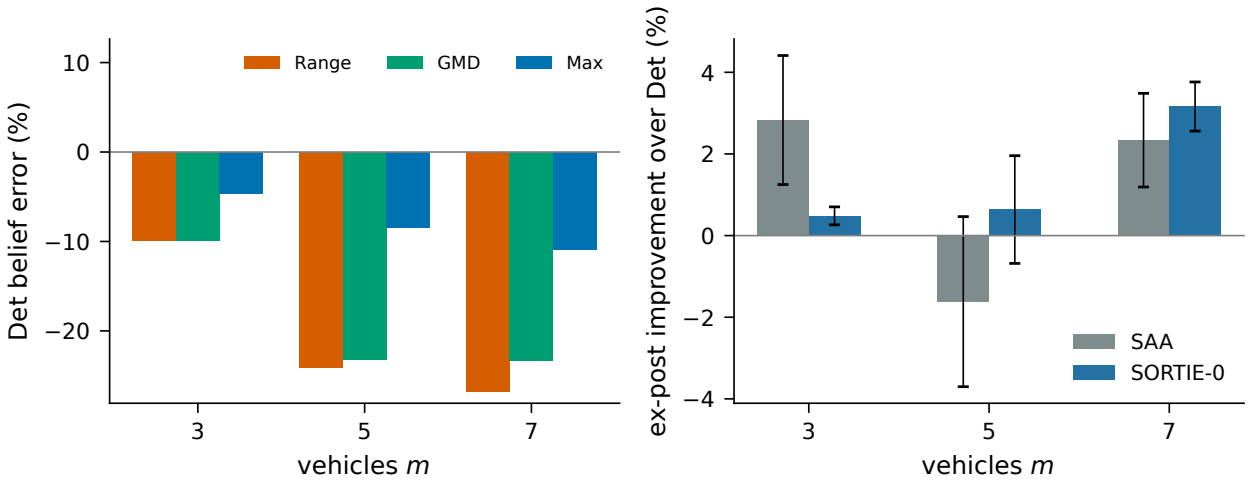


Figure 13: E10: belief error of Det and improvement over Det against the number of vehicles.

a lower level, both the deterministic and the stochastic planners produced detours in almost every range-optimised plan.

Optimising ex-post fairness improved the realised fairness by about one percent over a deterministic fair plan under homogeneous noise (up to 2% for the maximum), consistently but modestly, because balancing route lengths already roughly balances route risks when variance grows with length. When noise is heterogeneous, as with congestion concentrated in part of the network, the illusion grows to 26% for the range (23% on the same cells without congestion), and the gain of SORTIE grows only to about 1.5%. The improvement of the plan itself is significant but modest at the standard budget, and it grows where the deterministic model has most freedom to choose wrongly: with more slack in the budget (up to 7%, with a lower total length), with more vehicles (3% with seven) and on instances whose depot is not a block corner (2.2%, better on all twelve instances of the second family). It survives misspecified and spatially correlated noise, it is not reproduced by a risk-adjusted deterministic model, which is worse than the plain one, and a sample average needs thousands of scenarios to come close. On instances with 100 customers the faster search makes the improvement over the deterministic plan statistically significant when search noise is averaged

Table 13: E11: ratio of each measure with the day factor to its value without, mean ± standard deviation over the 500 E3 plans with $B = 1.10$.

s_Y	Max	Range	GMD	MAD	Var	CVaR90 range	$e^{s_Y^2}$
0.00	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000
0.15	1.000 ± 0.001	1.000 ± 0.001	1.000 ± 0.001	1.000 ± 0.001	1.023 ± 0.002	1.074 ± 0.016	1.023
0.30	1.000 ± 0.002	1.000 ± 0.002	1.000 ± 0.002	1.000 ± 0.002	1.095 ± 0.005	1.251 ± 0.042	1.094
0.50	1.001 ± 0.003	1.001 ± 0.003	1.001 ± 0.003	1.001 ± 0.003	1.285 ± 0.011	1.579 ± 0.081	1.284

Table 14: E12: noise-model misspecification. PBOI error of the gamma-based prediction against the true expected value under each family, and improvement of SORTIE-0 over Det evaluated under that family (format as Table 5).

Family	Source	Plans	PBOI |err| (%)	PBOI signed err. (%)	SORTIE-0 vs Det
Gamma	E3	500	0.16	+0.04	+1.23* (81/125)
Incident	E3	500	1.43	+1.35	+1.03* (71/125)
Lognormal	E3	500	0.29	+0.16	+1.21* (82/125)
Regional	E3	500	1.88	-0.74	+0.96* (65/125)
Gamma	E8	300	0.18	+0.10	+1.51* (53/75)
Lognormal	E8	300	0.51	+0.40	+1.48* (55/75)

over three seeds and the 18 instance–measure pairs are pooled (4%, $p = 0.02$), although no single measure is significant on six instances, and it does not separate SORTIE from a 200-scenario SAA at that size.

Limitations. The lower level is exact only for routes of up to 15 customers; beyond that it relies on multi-start local search (warm-started from the screening tour), and the bilevel constraint is then satisfied only approximately. PBOI assumes conditionally independent route durations; dependence beyond a common multiplicative factor (for example spatially correlated congestion affecting several routes) would need the copula or simulation layer we did not develop. At $n = 100$ single runs are still noisy and some are not converged after 240 s; the risk-consistent variant ($\kappa = 1$) is too slow to converge at $n \geq 75$; and at that size SORTIE could not be separated statistically from a 200-scenario SAA. The time limits are per run and are slightly exceeded by the final iteration. All instances are slices of one benchmark file with five vehicles.

Future work. Exact methods for the stochastic bilevel problem (branch-and-cut with PBOI-based bounds), correlated route durations, multi-day fairness, in which unfairness can be compensated across days, and time-dependent travel times.

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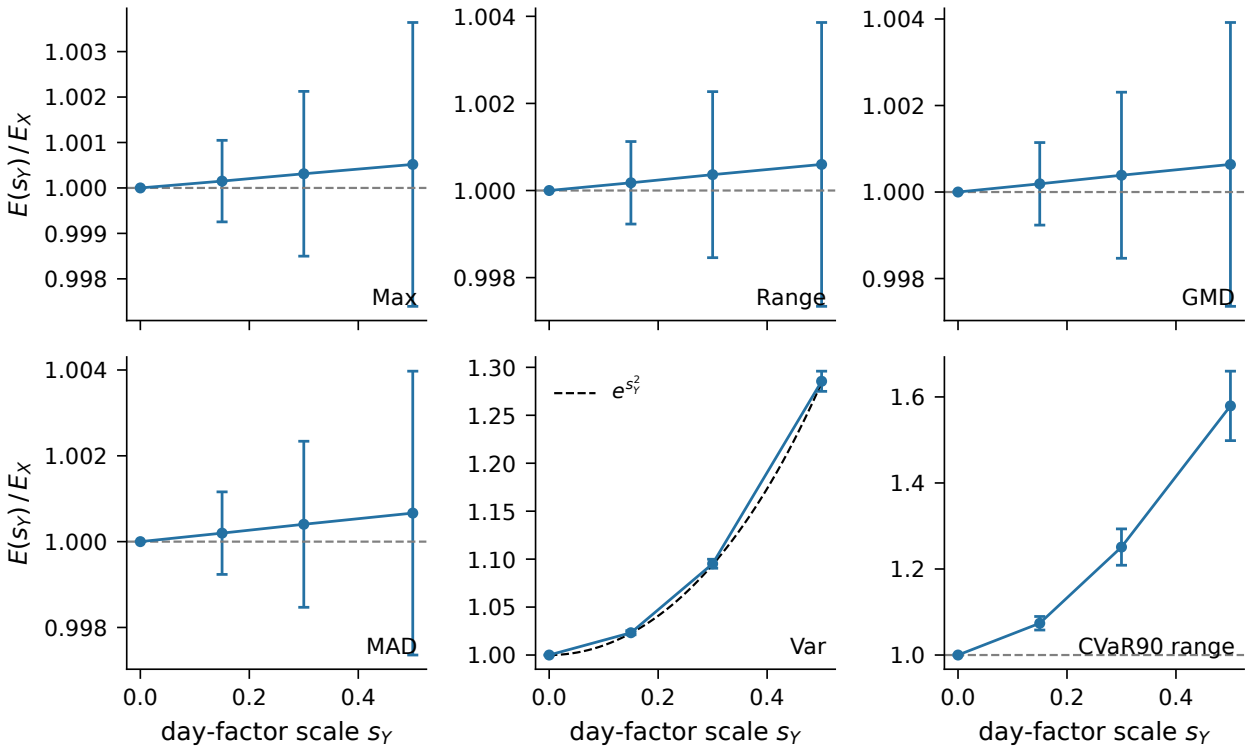


Figure 14: E11: effect of the day-factor dispersion s_Y on the expected ordered measures, the expected variance and the CVaR of the range.

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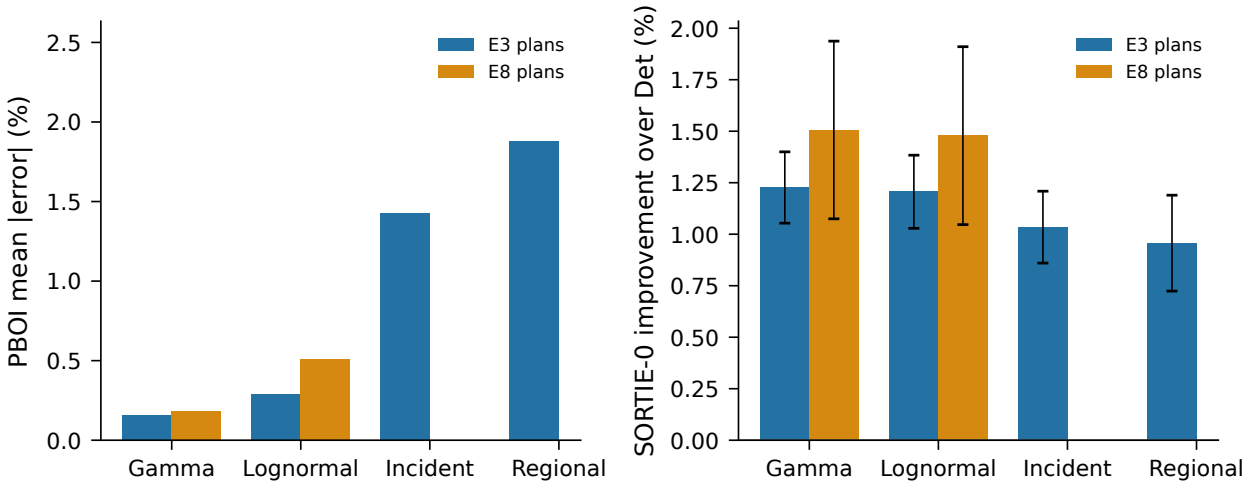


Figure 15: E12: PBOI error and improvement of SORTIE-0 over Det under four arc-noise families.

Table 15: E13: risk-consistent lower level, $n \in \{25, 35\}$, $B = 1.10$, ten instances per row. Routes changed: share of routes whose tour differs from the shortest tour of the same customers; extra length: total, per plan; mean risk: mean of $L_k + v\sqrt{W_k}$.

Measure	κ	Routes changed (%)	Extra length	Mean risk	Ex-post vs κ=0 [IQR]	Cost ratio
Range	0	0.0	0.0	1913.9	1.000 [1.000, 1.000]	1.098
Range	0.5	6.0	1.0	1912.9	1.000 [0.998, 1.000]	1.098
Range	1	4.0	0.9	1915.6	1.000 [0.995, 1.000]	1.098
Range	2	14.0	9.2	1914.0	1.000 [0.989, 1.004]	1.099
Range	4	14.0	14.1	1911.4	0.997 [0.989, 1.004]	1.099
Max	0	0.0	0.0	1890.7	1.000 [1.000, 1.000]	1.086
Max	0.5	2.0	0.5	1885.9	1.000 [1.000, 1.000]	1.084
Max	1	6.0	2.6	1891.3	1.000 [1.000, 1.000]	1.087
Max	2	6.0	3.9	1887.0	1.000 [1.000, 1.000]	1.085
Max	4	16.0	12.4	1882.6	1.000 [1.000, 1.000]	1.083

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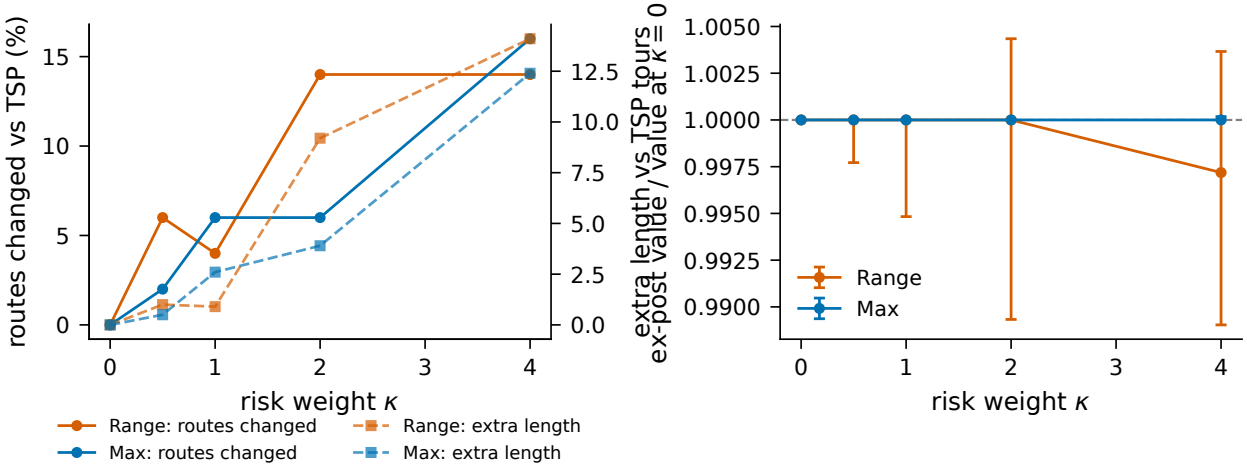


Figure 16: E13: effect of the drivers' risk weight κ.

Table 16: E14: rank-price prediction against exact evaluation. Groups: (instance, plan) pairs with at least one exactly improving move; top-8 hit: share of those groups in which the exact-best improving move is among the 8 best predicted moves (in brackets: expected share under a random order).

Measure	Plans	Moves	Groups with improving move	Spearman ρ	Top-8 hit (random)
Range	Cost plan	1448	10/10	0.93	0.80 (0.07)
Range	SORTIE-0 plan	346	1/10	0.80	0.00 (0.19)
Range	All plans	1794	11/20	0.92	0.73 (0.08)
GMD	Cost plan	1448	10/10	0.94	0.80 (0.07)
GMD	SORTIE-0 plan	333	0/10	0.78	– (–)
GMD	All plans	1781	10/20	0.94	0.80 (0.07)
Max	Cost plan	1448	10/10	0.89	0.80 (0.07)
Max	SORTIE-0 plan	554	1/10	0.89	0.00 (0.21)
Max	All plans	2002	11/20	0.89	0.73 (0.08)

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Table 17: E14: number of candidates K evaluated per step; ex-post value relative to $K = 8$, median and interquartile range over 20 instance–measure pairs.

K	Range median [IQR]	GMD median [IQR]
1	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
2	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
4	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
8	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
16	1.000 [1.000, 1.000]	1.000 [0.996, 1.000]
32	1.000 [0.994, 1.000]	1.000 [1.000, 1.000]

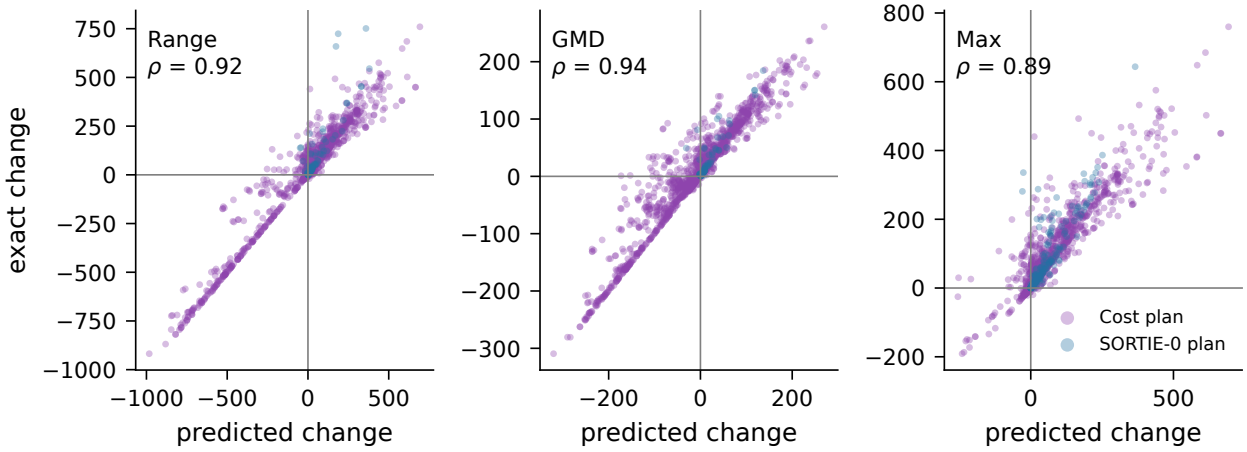


Figure 17: E14: predicted (rank prices) against exact change of the objective for 5577 moves.

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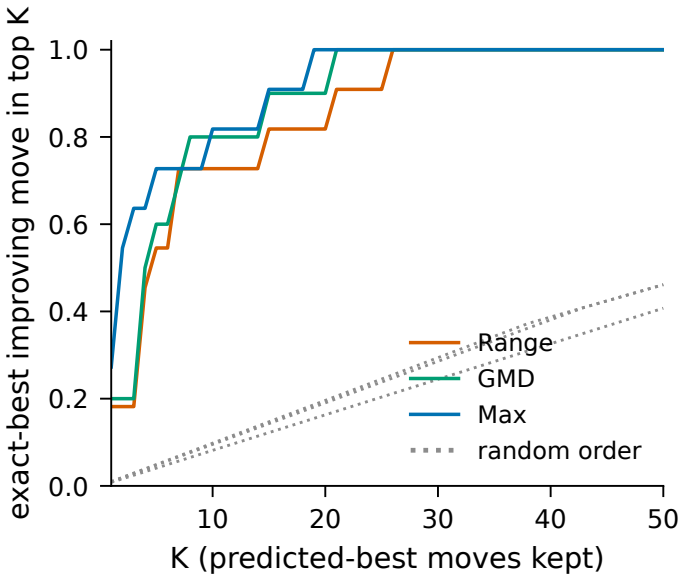


Figure 18: E14: share of cases in which the exact-best improving move is among the K best predicted moves.

Table 18: E15: improvement of SORTIE-0 over SAA with S scenarios, $n \in \{25, 35\}$, $B = 1.10$, ten instances per measure (format as Table 5; positive: SORTIE-0 better).

S	Range	GMD	Max	Pooled	p (pooled)
50	+0.17 (6/10)	+2.88* (8/10)	+0.99* (8/10)	+1.35* (22/30)	0.001
200	+2.11* (7/10)	+2.15* (7/10)	+0.46* (10/10)	+1.57* (24/30)	0.000
1000	+1.45 (5/10)	+1.64 (4/10)	+0.19 (6/10)	+1.09* (15/30)	0.002
5000	+1.34 (3/10)	+1.38 (2/10)	+0.02 (4/10)	+0.91 (9/30)	0.050

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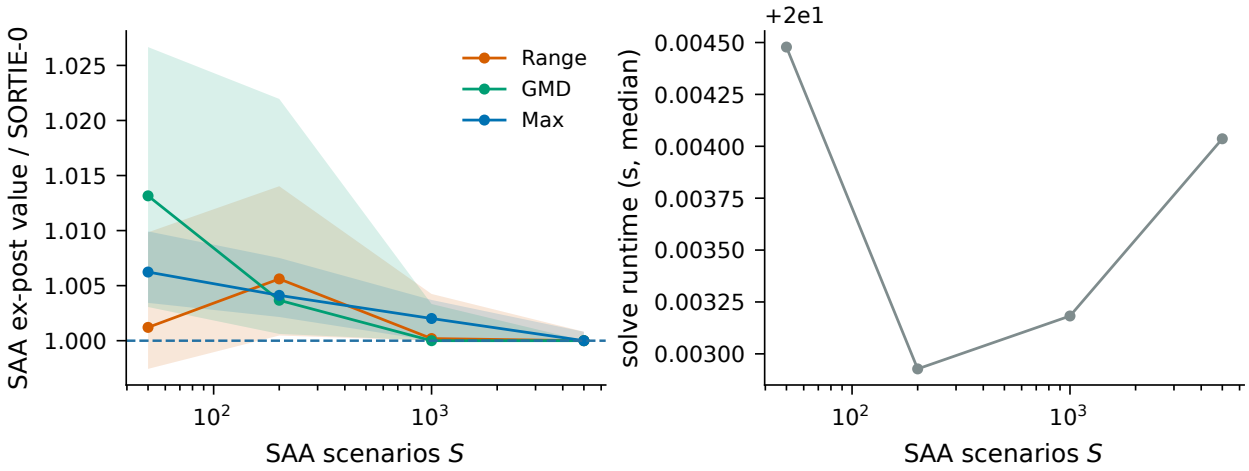


Figure 19: E15: ex-post value of SAA relative to SORTIE-0 against the sample size S.

Table 19: E16: risk-adjusted deterministic (Det-σ) and normal-marginal (PBOI-N) objectives, E3 grid, $B = 1.10$ (format as Table 5; the variance is omitted for PBOI-N because the expected empirical variance does not depend on the marginal family).

Measure	Det-σ vs Det	PBOI-N vs Det	SORTIE-0 vs Det	SORTIE-0 vs Det-σ
Range	-0.49 (8/25)	+0.48* (18/25)	+1.07* (17/25)	+1.51* (19/25)
GMD	-0.69 (8/25)	+0.83* (14/25)	+0.94* (14/25)	+1.50* (18/25)
Max	-0.08 (8/25)	+2.08* (24/25)	+2.08* (24/25)	+2.13* (25/25)
Pooled	-0.81† (37/125)	+1.10* (71/100)	+1.23* (81/125)	+1.86* (97/125)

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Table 20: E17: second benchmark family (12 X instances, depot plus first 50 customers), $B = 1.10$, 120 s, mean of two solver seeds; improvement of the ex-post value over Det (format as Table 5).

Measure	SAA vs Det	SORTIE-0 vs Det	Det-σ vs Det	Det belief err. (%)
Range	+1.48 (8/12)	+2.91* (11/12)	-0.39 (5/12)	-40.3
GMD	-0.62 (5/12)	+1.64 (8/12)	-2.15† (1/12)	-34.7
Max	+1.36* (10/12)	+1.93* (12/12)	+0.01 (7/12)	-8.4
Pooled	+0.74* (23/36)	+2.16* (31/36)	-0.84 (13/36)	-27.8

Table 21: E17 per instance: depot position, C^* and the improvement of SORTIE-0 over Det pooled over the three measures.

Instance	Depot	C^*	SORTIE-0 vs Det, mean of measures (wins/3)
X-n101-k25-h50	intermediate	7415	+2.80 (3/3)
X-n106-k14-h50	eccentric	9767	+2.02 (3/3)
X-n110-k13-h50	central	7194	+3.65 (3/3)
X-n115-k10-h50	central	7183	+2.04 (3/3)
X-n120-k6-h50	eccentric	9809	+1.89 (3/3)
X-n125-k30-h50	eccentric	9842	+1.58 (3/3)
X-n129-k18-h50	eccentric	9821	+1.03 (2/3)
X-n134-k13-h50	eccentric	4686	+0.69 (2/3)
X-n139-k10-h50	central	7253	+4.24 (3/3)
X-n143-k7-h50	eccentric	10398	+2.94 (2/3)
X-n148-k46-h50	intermediate	7752	+1.76 (2/3)
X-n153-k22-h50	intermediate	5786	+1.27 (2/3)

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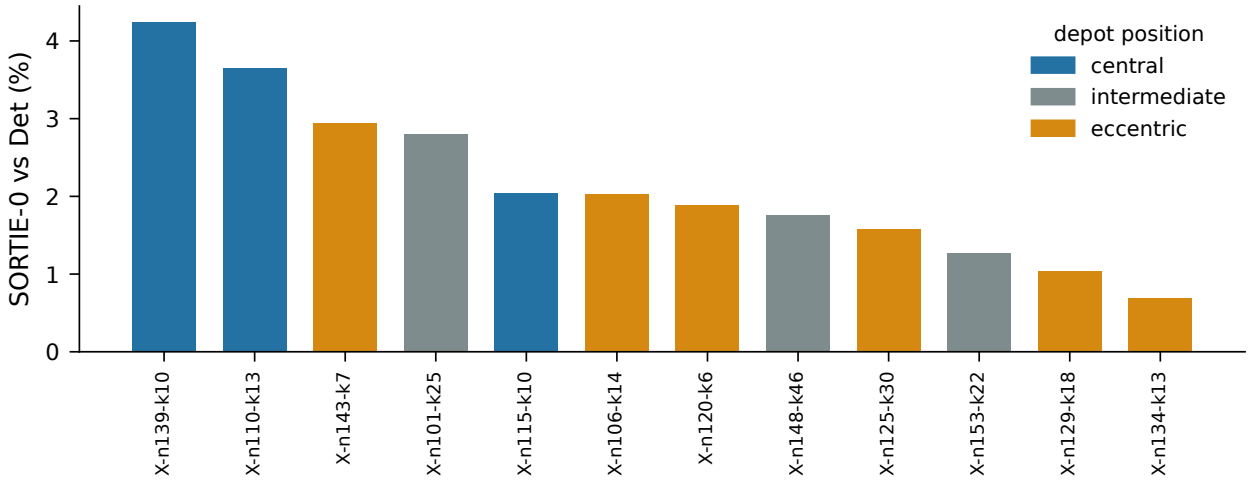


Figure 20: E17: improvement of SORTIE-0 over Det per instance (mean over measures and seeds), coloured by depot position.

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A Per-instance results

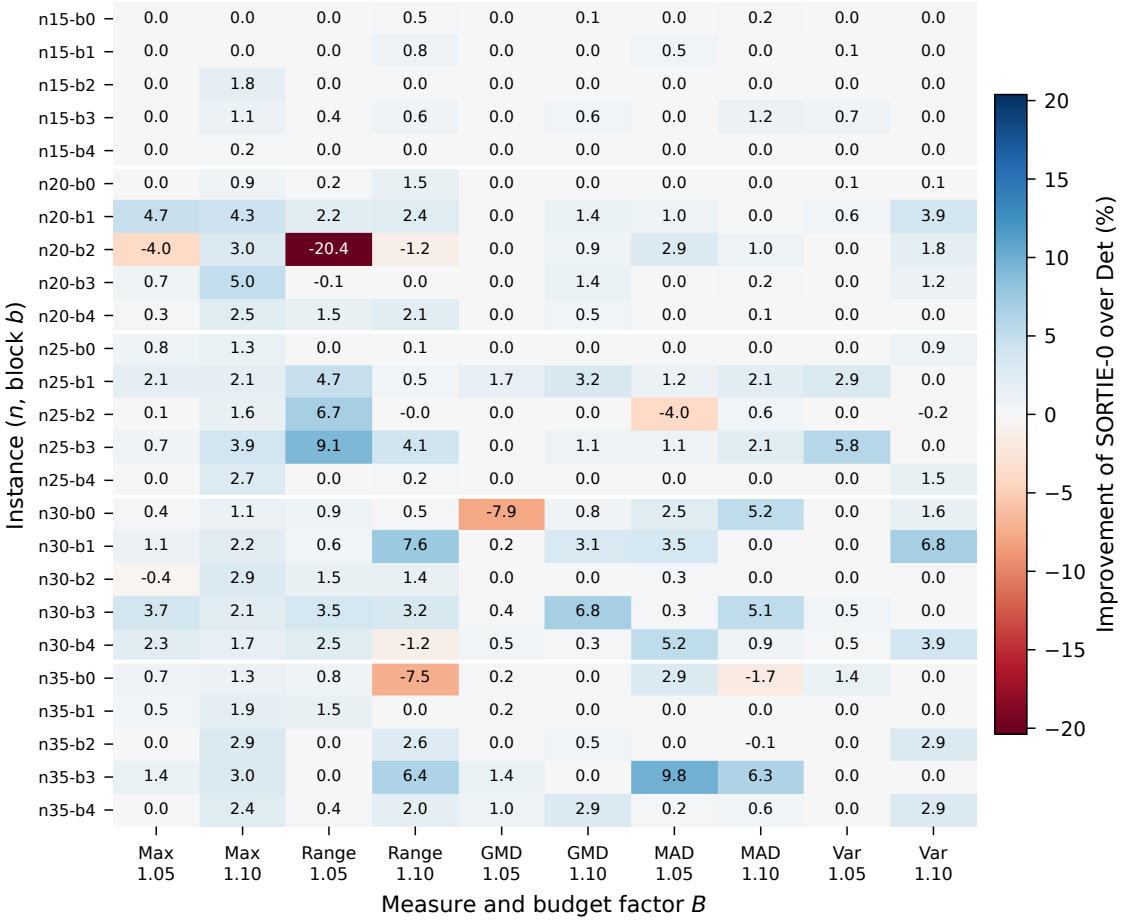


Figure 21: E3: improvement of the ex-post value of SORTIE-0 over Det (%) for every instance, measure and budget.

Tables 22 and 23 list the out-of-sample ex-post value of every E3 plan, Table 24 the seed means and standard deviations of E7b, and Table 25 the E17 values per instance and measure . Every plan is stored in the released plan cache and can be re-evaluated with the released simulator.

Table 22: E3, $B = 1.10$: out-of-sample ex-post value of every plan (bold: best); Cost: the cost-optimal plan; S-0/S-1: SORTIE-0/1; Gain: improvement of S-0 over Det; last column: total length of the S-0 plan relative to C^*.

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
n15-b0	Max	2,640	2,547	2,561	2,547	**2,545**	+0.00	1.026
	Range	2,412	1,575	**1,568**	**1,568**	1,573	+0.47	1.098
	GMD	907	625	625	625	**623**	+0.06	1.100
	MAD	652	468	**467**	**467**	467	+0.20	1.100
	Var	730,909	357,317	357,317	357,238	**354,022**	+0.02	1.100
n15-b1	Max	2,165	2,091	2,095	**2,091**	**2,091**	+0.00	1.098

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
	Range	1,875	1,295	**1,285**	**1,285**	**1,285**	+0.79	1.095
	GMD	729	**530**	**530**	**530**	**530**	+0.00	1.095
	MAD	537	**405**	**405**	**405**	**405**	+0.00	1.095
	Var	461,317	**258,741**	261,374	**258,741**	**258,741**	+0.00	1.095
n15-b2	Max	2,419	2,419	2,376	**2,376**	**2,376**	+1.78	1.070
	Range	2,187	**1,231**	1,234	**1,231**	**1,231**	+0.00	1.081
	GMD	804	**478**	482	**478**	479	+0.00	1.099
	MAD	568	**347**	**347**	**347**	348	+0.00	1.099
	Var	604,666	**222,814**	223,451	**222,814**	223,451	+0.00	1.099
n15-b3	Max	1,926	1,826	1,826	1,805	**1,803**	+1.13	1.070
	Range	1,899	784	806	779	**773**	+0.64	1.070
	GMD	665	305	305	303	**300**	+0.59	1.070
	MAD	452	226	224	223	**221**	+1.15	1.070
	Var	471,206	90,921	95,342	90,921	**89,222**	+0.00	1.095
n15-b4	Max	2,412	1,940	**1,934**	1,935	1,935	+0.25	1.056
	Range	2,170	**1,024**	**1,024**	**1,024**	**1,024**	+0.00	1.098
	GMD	772	**417**	**417**	**417**	**417**	+0.00	1.098
	MAD	531	**316**	**316**	**316**	**316**	+0.00	1.098
	Var	543,628	**164,264**	**164,264**	**164,264**	**164,264**	+0.00	1.098
n20-b0	Max	2,656	2,651	2,647	**2,628**	**2,628**	+0.90	1.081
	Range	2,428	1,549	1,537	1,526	**1,522**	+1.48	1.097
	GMD	926	628	631	628	**626**	+0.00	1.099
	MAD	672	477	481	477	**477**	+0.00	1.099
	Var	782,670	358,873	361,706	358,394	**356,722**	+0.13	1.097
n20-b1	Max	1,756	1,743	**1,668**	1,669	1,669	+4.28	1.037
	Range	1,332	563	562	550	**550**	+2.42	1.099
	GMD	472	217	217	214	**214**	+1.40	1.077
	MAD	323	157	159	**157**	**157**	+0.04	1.077
	Var	230,329	47,975	47,125	**46,117**	**46,117**	+3.87	1.077
n20-b2	Max	2,393	2,224	2,177	**2,158**	**2,158**	+2.97	1.096
	Range	2,309	**1,283**	**1,283**	1,299	1,293	-1.24	1.090
	GMD	886	558	**552**	553	553	+0.92	1.099
	MAD	645	387	**383**	383	383	+1.02	1.099
	Var	692,985	257,489	**252,913**	**252,913**	**252,913**	+1.78	1.098
n20-b3	Max	2,668	2,472	2,349	**2,348**	**2,348**	+5.02	1.084
	Range	2,518	**1,272**	1,278	**1,272**	**1,272**	+0.00	1.100
	GMD	918	523	517	**516**	523	+1.42	1.099
	MAD	644	373	**372**	**372**	374	+0.17	1.099
	Var	757,430	250,084	247,116	247,116	**245,946**	+1.19	1.098
n20-b4	Max	1,981	1,823	1,782	1,778	**1,776**	+2.48	1.068
	Range	1,472	1,085	1,066	1,062	**1,060**	+2.13	1.068
	GMD	617	399	398	397	**396**	+0.52	1.067
	MAD	477	282	**282**	**282**	**282**	+0.14	1.098
	Var	332,502	147,358	148,357	147,358	**146,846**	+0.00	1.067
n25-b0	Max	2,607	2,622	2,613	**2,588**	**2,588**	+1.30	1.095
	Range	2,379	1,253	1,260	**1,252**	1,253	+0.11	1.100
	GMD	858	**506**	508	**506**	507	+0.00	1.100
	MAD	597	381	383	381	**381**	+0.00	1.100
	Var	718,676	245,392	248,480	**243,295**	**243,295**	+0.85	1.099
n25-b1	Max	2,145	1,971	1,939	**1,931**	**1,931**	+2.05	1.027
	Range	1,257	618	615	615	**611**	+0.54	1.098
	GMD	535	247	238	239	**236**	+3.19	1.098

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
	MAD	417	179	176	175	**173**	+2.08	1.098
	Var	255,701	58,275	60,272	58,275	**57,003**	+0.00	1.098
n25-b2	Max	2,148	2,065	2,035	**2,033**	**2,033**	+1.56	1.097
	Range	2,036	**1,312**	1,312	1,312	1,312	-0.01	1.099
	GMD	763	539	539	539	**537**	+0.00	1.092
	MAD	546	412	409	409	**408**	+0.62	1.092
	Var	516,255	265,863	267,817	266,385	**264,172**	-0.20	1.099
n25-b3	Max	1,984	2,025	1,958	**1,946**	**1,946**	+3.90	1.097
	Range	1,474	874	840	838	**836**	+4.13	1.099
	GMD	560	331	333	**328**	331	+1.06	1.098
	MAD	405	248	244	**242**	**242**	+2.15	1.098
	Var	283,090	**104,507**	107,279	**104,507**	105,353	+0.00	1.098
n25-b4	Max	2,151	2,197	2,143	2,137	**2,137**	+2.74	1.099
	Range	1,931	1,002	1,004	999	**994**	+0.21	1.099
	GMD	683	393	394	393	**391**	+0.00	1.100
	MAD	468	291	292	291	**290**	+0.00	1.100
	Var	459,013	152,985	153,860	150,758	**149,397**	+1.46	1.099
n30-b0	Max	2,773	2,615	2,587	**2,585**	**2,585**	+1.14	1.050
	Range	2,626	1,229	1,228	1,223	**1,223**	+0.47	1.099
	GMD	1,000	497	497	**493**	493	+0.79	1.099
	MAD	724	392	372	**371**	372	+5.22	1.099
	Var	911,746	235,555	233,126	**231,861**	**231,861**	+1.57	1.099
n30-b1	Max	2,657	2,574	2,540	**2,516**	2,521	+2.23	1.075
	Range	2,287	1,537	1,421	**1,420**	**1,420**	+7.63	1.095
	GMD	913	568	551	551	**550**	+3.06	1.100
	MAD	684	411	**409**	411	412	+0.00	1.098
	Var	730,545	297,591	298,366	277,261	**275,214**	+6.83	1.095
n30-b2	Max	1,957	1,838	1,784	1,784	**1,783**	+2.94	1.078
	Range	1,838	1,002	988	988	**961**	+1.44	1.091
	GMD	667	365	372	365	**365**	+0.00	1.091
	MAD	466	259	263	259	**259**	+0.00	1.091
	Var	419,940	126,369	126,369	126,369	**126,081**	+0.00	1.091
n30-b3	Max	2,401	2,350	2,307	**2,300**	**2,300**	+2.13	1.097
	Range	2,371	1,264	**1,223**	**1,223**	1,284	+3.25	1.100
	GMD	853	532	502	**496**	500	+6.75	1.100
	MAD	592	396	**369**	375	**369**	+5.12	1.100
	Var	717,888	**225,491**	229,750	**225,491**	228,530	+0.00	1.100
n30-b4	Max	2,325	2,231	2,196	2,193	**2,191**	+1.69	1.068
	Range	2,100	**1,217**	1,237	1,232	1,231	-1.21	1.093
	GMD	762	494	502	493	**492**	+0.27	1.098
	MAD	533	347	351	344	**344**	+0.90	1.100
	Var	536,206	225,051	219,695	**216,280**	225,293	+3.90	1.099
n35-b0	Max	2,892	2,717	2,704	**2,682**	**2,682**	+1.27	1.098
	Range	2,744	**1,376**	1,741	1,479	1,724	-7.46	1.096
	GMD	1,013	**529**	609	**529**	609	+0.00	1.100
	MAD	718	421	429	428	**385**	-1.72	1.098
	Var	964,248	**259,530**	372,532	**259,530**	368,171	+0.00	1.100
n35-b1	Max	2,543	2,127	2,090	**2,087**	**2,087**	+1.91	1.096
	Range	2,429	**825**	**825**	**825**	**825**	+0.00	1.098
	GMD	946	**323**	324	**323**	**323**	+0.00	1.098
	MAD	696	**238**	239	**238**	**238**	+0.00	1.098
	Var	758,306	**102,506**	102,787	**102,506**	**102,506**	+0.00	1.098

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
n35-b2	Max	1,922	1,794	1,742	**1,741**	1,814	+2.91	1.061
	Range	1,526	553	546	**538**	**538**	+2.61	1.093
	GMD	562	209	**208**	**208**	**208**	+0.48	1.093
	MAD	398	**153**	153	153	153	-0.06	1.093
	Var	296,372	43,908	43,424	**42,644**	**42,644**	+2.88	1.095
n35-b3	Max	2,527	2,452	2,398	2,379	**2,379**	+2.98	1.100
	Range	2,137	1,066	1,012	**998**	**998**	+6.37	1.100
	GMD	780	**377**	381	**377**	**377**	+0.00	1.100
	MAD	548	291	273	273	**273**	+6.35	1.100
	Var	555,805	**142,628**	144,945	**142,628**	**142,628**	+0.00	1.100
n35-b4	Max	2,510	2,186	2,142	**2,133**	**2,133**	+2.43	1.085
	Range	2,436	702	702	**687**	**687**	+2.03	1.097
	GMD	902	272	275	**264**	**264**	+2.92	1.097
	MAD	640	194	**193**	**193**	**193**	+0.58	1.097
	Var	772,452	71,590	72,810	**69,523**	**69,523**	+2.89	1.097

Table 23: E3, $B = 1.05$: out-of-sample ex-post value of every plan (bold: best); Cost: the cost-optimal plan; S-0/S-1: SORTIE-0/1; Gain: improvement of S-0 over Det; last column: total length of the S-0 plan relative to C^*.

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
n15-b0	Max	2,640	**2,547**	2,559	**2,547**	**2,547**	+0.00	1.026
	Range	2,412	**2,319**	2,331	**2,319**	**2,319**	+0.00	1.026
	GMD	907	**868**	873	**868**	**868**	+0.00	1.026
	MAD	652	**621**	**621**	**621**	**621**	+0.00	1.026
	Var	730,909	**691,734**	699,905	**691,734**	**691,734**	+0.00	1.026
n15-b1	Max	2,165	**2,091**	**2,091**	**2,091**	**2,091**	+0.00	1.041
	Range	1,875	**1,503**	**1,503**	**1,503**	1,505	+0.00	1.040
	GMD	729	**616**	**616**	**616**	617	+0.00	1.040
	MAD	537	469	469	**467**	468	+0.50	1.048
	Var	461,317	337,238	339,053	**336,907**	337,815	+0.10	1.048
n15-b2	Max	2,419	**2,419**	**2,419**	**2,419**	**2,419**	+0.00	1.000
	Range	2,187	**2,187**	**2,187**	**2,187**	**2,187**	+0.00	1.000
	GMD	804	**804**	**804**	**804**	**804**	+0.00	1.000
	MAD	568	**568**	**568**	**568**	**568**	+0.00	1.000
	Var	604,666	**604,666**	**604,666**	**604,666**	**604,666**	+0.00	1.000
n15-b3	Max	1,926	**1,926**	**1,926**	**1,926**	**1,926**	+0.00	1.000
	Range	1,899	1,181	**1,177**	**1,177**	**1,177**	+0.37	1.043
	GMD	665	468	**468**	468	468	+0.00	1.041
	MAD	452	**349**	349	**349**	**349**	+0.00	1.041
	Var	471,206	200,220	**198,775**	**198,775**	**198,775**	+0.72	1.043
n15-b4	Max	2,412	**1,940**	**1,940**	**1,940**	**1,940**	+0.00	1.000
	Range	2,170	**1,698**	**1,698**	**1,698**	**1,698**	+0.00	1.000
	GMD	772	**648**	**648**	**648**	**648**	+0.00	1.022
	MAD	531	**466**	**466**	**466**	**466**	+0.00	1.022
	Var	543,628	**375,246**	**375,246**	**375,246**	**375,246**	+0.00	1.000
n20-b0	Max	2,656	**2,639**	2,647	**2,639**	**2,639**	+0.00	1.021
	Range	2,428	2,411	2,418	**2,405**	**2,405**	+0.25	1.022
	GMD	926	**863**	**863**	**863**	**863**	+0.00	1.049

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
	MAD	672	**595**	**595**	**595**	**595**	+0.00	1.049
	Var	782,670	748,918	751,999	**748,239**	**748,239**	+0.09	1.022
n20-b1	Max	1,756	1,751	**1,668**	1,669	1,669	+4.72	1.037
	Range	1,332	666	**652**	**652**	**652**	+2.20	1.036
	GMD	472	251	**251**	251	251	+0.00	1.049
	MAD	323	185	184	**183**	**183**	+0.99	1.049
	Var	230,329	62,141	**61,759**	**61,759**	**61,759**	+0.62	1.036
n20-b2	Max	2,393	**2,191**	2,202	2,279	2,279	-4.03	1.039
	Range	2,309	**1,652**	1,988	1,988	1,988	-20.38	1.039
	GMD	886	**656**	**656**	**656**	**656**	+0.00	1.049
	MAD	645	505	**490**	**490**	**490**	+2.92	1.049
	Var	692,985	**379,751**	**379,751**	**379,751**	**379,751**	+0.00	1.049
n20-b3	Max	2,668	2,398	**2,381**	**2,381**	**2,381**	+0.71	1.046
	Range	2,518	**1,640**	1,642	1,642	1,642	-0.10	1.044
	GMD	918	**673**	**673**	**673**	**673**	+0.00	1.046
	MAD	644	**490**	**490**	**490**	**490**	+0.00	1.046
	Var	757,430	**408,393**	**408,393**	**408,393**	**408,393**	+0.00	1.046
n20-b4	Max	1,981	1,819	**1,814**	**1,814**	1,815	+0.30	1.050
	Range	1,472	1,235	**1,216**	**1,216**	**1,216**	+1.51	1.047
	GMD	617	**485**	**485**	**485**	**485**	+0.00	1.048
	MAD	477	**341**	**341**	**341**	**341**	+0.00	1.048
	Var	332,502	208,876	208,876	208,876	**208,721**	+0.00	1.045
n25-b0	Max	2,607	2,610	2,613	**2,589**	**2,589**	+0.81	1.019
	Range	2,379	**1,755**	**1,755**	**1,755**	**1,755**	+0.00	1.050
	GMD	858	**683**	**683**	**683**	**683**	+0.00	1.050
	MAD	597	**503**	**503**	**503**	**503**	+0.00	1.050
	Var	718,676	**416,021**	**416,021**	**416,021**	**416,021**	+0.00	1.050
n25-b1	Max	2,145	1,971	1,938	**1,931**	**1,931**	+2.05	1.027
	Range	1,257	774	745	**738**	**738**	+4.66	1.027
	GMD	535	290	288	**285**	**285**	+1.69	1.027
	MAD	417	212	211	**209**	**209**	+1.22	1.027
	Var	255,701	82,851	81,533	**80,430**	**80,430**	+2.92	1.027
n25-b2	Max	2,148	2,148	2,147	**2,145**	**2,145**	+0.13	1.029
	Range	2,036	1,815	**1,693**	**1,693**	**1,693**	+6.71	1.046
	GMD	763	**654**	**654**	**654**	689	+0.00	1.046
	MAD	546	**479**	504	498	**479**	-3.99	1.040
	Var	516,255	418,916	**372,661**	418,916	417,741	+0.00	1.036
n25-b3	Max	1,984	1,968	1,958	**1,953**	**1,953**	+0.72	1.045
	Range	1,474	1,162	1,058	**1,056**	**1,056**	+9.13	1.045
	GMD	560	**419**	423	**419**	**419**	+0.00	1.046
	MAD	405	314	312	**311**	**311**	+1.10	1.050
	Var	283,090	169,897	**160,109**	**160,109**	**160,109**	+5.76	1.046
n25-b4	Max	2,151	2,151	2,151	**2,150**	**2,150**	+0.03	1.044
	Range	1,931	**1,433**	1,544	**1,433**	1,544	+0.00	1.034
	GMD	683	**565**	**565**	**565**	**565**	+0.00	1.044
	MAD	468	**397**	**397**	**397**	**397**	+0.00	1.044
	Var	459,013	**294,618**	**294,618**	**294,618**	**294,618**	+0.00	1.044
n30-b0	Max	2,773	2,606	2,602	**2,597**	2,597	+0.36	1.044
	Range	2,626	2,388	2,393	2,368	**2,357**	+0.86	1.044
	GMD	1,000	**760**	831	820	821	-7.91	1.044
	MAD	724	566	551	552	**548**	+2.54	1.044
	Var	911,746	516,782	516,782	516,782	**515,564**	+0.00	1.048

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
n30-b1	Max	2,657	2,552	2,530	**2,524**	2,526	+1.13	1.037
	Range	2,287	1,903	1,893	**1,892**	1,893	+0.61	1.041
	GMD	913	759	761	758	**706**	+0.16	1.044
	MAD	684	535	516	**516**	517	+3.54	1.049
	Var	730,545	**460,952**	**460,952**	**460,952**	**460,952**	+0.00	1.044
n30-b2	Max	1,957	**1,942**	1,958	1,950	1,949	-0.39	1.034
	Range	1,838	1,224	1,217	1,205	**1,204**	+1.53	1.049
	GMD	667	508	514	508	**508**	+0.00	1.048
	MAD	466	393	393	**392**	394	+0.28	1.049
	Var	419,940	225,689	230,414	225,689	**225,408**	+0.00	1.048
n30-b3	Max	2,401	2,387	2,306	**2,300**	**2,300**	+3.67	1.042
	Range	2,371	1,946	**1,878**	**1,878**	**1,878**	+3.51	1.050
	GMD	853	664	**661**	**661**	**661**	+0.39	1.050
	MAD	592	453	453	**451**	**451**	+0.32	1.050
	Var	717,888	449,606	**447,388**	**447,388**	**447,388**	+0.49	1.050
n30-b4	Max	2,325	2,244	2,200	**2,193**	**2,193**	+2.28	1.045
	Range	2,100	1,739	1,701	**1,695**	**1,695**	+2.53	1.043
	GMD	762	632	632	**629**	630	+0.51	1.050
	MAD	533	457	**433**	**433**	**433**	+5.16	1.048
	Var	536,206	354,473	364,184	**352,875**	**352,875**	+0.45	1.046
n35-b0	Max	2,892	2,717	2,703	2,697	**2,694**	+0.75	1.035
	Range	2,744	2,488	2,486	2,468	**2,447**	+0.80	1.035
	GMD	1,013	846	860	844	**841**	+0.24	1.028
	MAD	718	580	574	564	**562**	+2.88	1.028
	Var	964,248	809,215	800,421	798,285	**793,815**	+1.35	1.022
n35-b1	Max	2,543	2,245	**2,137**	2,233	**2,137**	+0.52	1.045
	Range	2,429	1,793	1,768	1,767	**1,239**	+1.48	1.049
	GMD	946	502	502	501	**499**	+0.22	1.050
	MAD	696	**374**	**374**	**374**	**374**	+0.00	1.050
	Var	758,306	228,342	230,146	228,342	**228,317**	+0.00	1.050
n35-b2	Max	1,922	**1,748**	**1,748**	**1,748**	**1,748**	+0.00	1.047
	Range	1,526	**620**	**620**	**620**	**620**	+0.00	1.047
	GMD	562	**246**	**246**	**246**	**246**	+0.00	1.047
	MAD	398	**184**	**184**	**184**	**184**	+0.00	1.047
	Var	296,372	**58,746**	**58,746**	**58,746**	**58,746**	+0.00	1.047
n35-b3	Max	2,527	2,453	2,420	**2,420**	2,423	+1.36	1.037
	Range	2,137	**1,292**	**1,292**	**1,292**	**1,292**	+0.00	1.050
	GMD	780	532	**524**	**524**	**524**	+1.42	1.050
	MAD	548	440	**397**	**397**	**397**	+9.77	1.050
	Var	555,805	**247,798**	**247,798**	**247,798**	**247,798**	+0.00	1.050
n35-b4	Max	2,510	**2,153**	2,174	**2,153**	**2,153**	+0.00	1.025
	Range	2,436	1,026	**1,022**	**1,022**	**1,022**	+0.38	1.050
	GMD	902	415	**411**	**411**	**411**	+0.96	1.050
	MAD	640	290	**290**	**290**	**290**	+0.24	1.048
	Var	772,452	**156,368**	158,633	**156,368**	158,899	+0.00	1.050

Table 24: E7b, $n = 100$: mean (standard deviation) over three solver seeds of the ex-post value.

Instance	Measure	Det	SAA	S-0	S-0 vs Det (%)
n100-b0	GMD	378 (53.9)	349 (11.7)	337 (3.7)	+10.77
n100-b0	Max	3,016 (137)	2,930 (6.2)	2,924 (6.3)	+3.06
n100-b0	Range	870 (1.0)	861 (12.9)	840 (0.3)	+3.44
n100-b1	GMD	249 (30.4)	231 (1.6)	235 (4.6)	+5.71
n100-b1	Max	2,727 (9.7)	2,665 (2.7)	2,667 (0.3)	+2.20
n100-b1	Range	604 (3.5)	648 (27.8)	610 (8.6)	-1.05
n100-b2	GMD	346 (17.6)	351 (6.7)	401 (39.8)	-15.87
n100-b2	Max	2,444 (8.5)	2,493 (32.0)	2,484 (62.5)	-1.65
n100-b2	Range	1,081 (318)	929 (44.4)	970 (5.4)	+10.26
n100-b3	GMD	304 (10.3)	313 (14.9)	305 (7.8)	-0.20
n100-b3	Max	2,583 (51.7)	2,535 (0.2)	2,602 (117)	-0.72
n100-b3	Range	881 (10.6)	785 (7.3)	799 (9.8)	+9.35
n100-b4	GMD	516 (5.5)	523 (19.8)	502 (11.3)	+2.79
n100-b4	Max	2,525 (8.6)	2,497 (12.4)	2,475 (15.8)	+2.02
n100-b4	Range	1,346 (64.1)	1,365 (63.8)	1,335 (50.6)	+0.83
n100-b5	GMD	289 (6.6)	273 (4.1)	243 (22.8)	+15.89
n100-b5	Max	2,396 (12.8)	2,431 (62.3)	2,391 (8.5)	+0.23
n100-b5	Range	890 (282)	700 (85.3)	630 (69.9)	+29.24

Table 25: E17: ex-post value, mean over two solver seeds (bold: best).

Instance	Measure	Det	SAA	S-0	Det-σ	S-0 vs Det (%)
X-n101-k25-h50	GMD	267	268	**265**	279	+0.65
	Max	2,049	2,027	**2,013**	2,060	+1.73
	Range	755	731	**709**	767	+6.01
X-n106-k14-h50	GMD	499	508	**497**	513	+0.36
	Max	2,874	2,821	**2,811**	2,869	+2.19
	Range	1,317	1,285	**1,271**	1,300	+3.52
X-n110-k13-h50	GMD	154	152	**148**	156	+4.21
	Max	1,781	1,731	**1,728**	1,751	+2.97
	Range	397	392	**382**	395	+3.76
X-n115-k10-h50	GMD	167	160	**159**	174	+4.67
	Max	1,788	1,790	**1,781**	1,800	+0.40
	Range	409	415	**405**	414	+1.07
X-n120-k6-h50	GMD	609	662	**583**	636	+4.32
	Max	2,843	2,830	**2,829**	2,859	+0.51
	Range	1,493	1,499	**1,480**	1,518	+0.83
X-n125-k30-h50	GMD	358	359	**352**	365	+1.69
	Max	2,689	2,670	**2,642**	2,696	+1.75
	Range	929	929	916	**914**	+1.30
X-n129-k18-h50	GMD	**557**	581	565	560	-1.31
	Max	2,843	2,817	**2,795**	2,842	+1.69
	Range	1,437	1,399	**1,398**	1,417	+2.72
X-n134-k13-h50	GMD	460	**460**	460	463	+0.05
	Max	1,935	1,936	**1,894**	1,986	+2.08
	Range	**1,354**	1,358	1,355	1,378	-0.06

Instance	Measure	Det	SAA	S-0	Det-σ	S-0 vs Det (%)
X-n139-k10-h50	GMD	173	160	**159**	164	+8.01
	Max	1,804	1,788	**1,785**	1,792	+1.07
	Range	428	413	**412**	432	+3.63
X-n143-k7-h50	GMD	536	**535**	544	550	-1.52
	Max	3,087	2,979	**2,960**	3,038	+4.10
	Range	1,442	1,365	**1,352**	1,482	+6.24
X-n148-k46-h50	GMD	**250**	252	253	258	-0.95
	Max	2,087	2,021	**2,017**	2,080	+3.33
	Range	600	597	**583**	586	+2.88
X-n153-k22-h50	GMD	**177**	186	178	188	-0.54
	Max	1,523	1,506	**1,504**	1,517	+1.28
	Range	472	466	**457**	480	+3.06